

Title: Tone diachrony beyond tonogenesis: Reconstructing the tone melodies of Proto-Edoid (Niger-Congo)

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Abstract: The immediate goal of this paper is to reconstruct nominal tone in the Edoid family using the comparative method, one branch within the Benue-Congo subfamily of the Niger-Congo phylum, a stock for which no original tonogenesis event is discernible. To this end, we develop the Proto-Edoid Tone (PEdT) database drawn from 40 sources on 21 modern Edoid languages, totaling over 3000 individual data points, from which 436 cognates can be identified. Based on the PEdT database, we systematically compare tone patterns across the four proposed branches of Edoid, which results in a number of tonal correspondences which we reconstruct as combinations of tone on a noun class prefix and tone on a nominal stem. We call these nominal tone melodies. Based on these intermediate reconstructions, we reconstruct tone in Proto-Edoid as a symmetrical set of melodies $**L-L$, $**L-LH$, $**L-H$, $**L-HL$, $**H-L$, $**H-LH$, $**H-HL$, and $**H-H$. From these Proto-Edoid melodies, we track several recurring tone changes, the most common of which are Rightward H-Shift (e.g. the Great Edo Tone Shift, $*H-L > L-H$), H-Obligatoriness (i.e. adding a H to all-L words), and H-Lowering (i.e. $*H > {}^{\downarrow}H$, M, or L). These tone changes show both the hallmarks of regular sound change as well as many unexpected reflexes which cannot be explained at this point, setting the stage for future work on Edoid tone and how it relates to Proto-Benue-Congo and ultimately Proto-Niger-Congo.

Keywords: tone, tonal reconstruction, tone change, comparative method, Africa, Edoid

1 Introduction

In the diachronic tone literature, tonogenesis events involving the “transphonologization” of a segmental contrast to a tonal one are widely studied (Kirby, Pittayaporn, & Brunelle 2022 – see

within for many references). Much less focus has been placed on how tone systems evolve after tonogenesis, and what the implications are for the study sound change. Specifically, how far back can tone systems be reconstructed in language families without any clear tonogenesis event? This is a question which hovers over comparative Africanist linguistics, where the great majority of the continent's languages are tonal with no indication that this is a recent phenomenon. Thus, it is generally assumed that the massive Niger-Congo phylum had tone even in its earliest proto-stage (Hyman 2020) rendering any original tonogenetic event irrecoverable, and that Niger-Congo languages without tone are innovative (e.g. famously in Swahili). In fact, evidence for any original tonogenetic event in any African family is remarkably lacking compared to other tonal areas of the world (Sæbø, Grossman, & Moran 2025).¹

Occasionally, dedicated studies have sought to specifically reconstruct the tone and tone patterns for individual African families using the methodologies of historical linguistics, e.g. Proto-Bantu tone (Greenberg 1948), Proto-Gbaya (Moñino 1995), Proto-Sara-Bongo-Bagirmi (Boyeldieu 2000), Proto-Khoe (Elderkin 2008), *inter alia*. Still, tone in the vast majority of families remains without a rigorous application of the comparative method. This paper represents one such attempt, with the immediate goal of reconstructing tone in the Edoid family, one branch within Benue-Congo, itself the largest branch within Niger-Congo (the largest phylum in Africa). To this end, we have created a lexical database on Edoid languages as part of the Proto-Edoid Tone Project (or “PEdT Project”), consisting of 3,038 individual data points in 21 modern languages drawn from 40 sources (included in the supplemental materials). Altogether, 436 cognates are

¹ That is, evidence for “strict tonogenesis” in Africa is lacking, whereby a non-tonal language acquires a tonal contrast for the first time (Sæbø, Grossman, & Moran 2025; see also Hyslop 2022).

identified based on Elugbe's (1989a) sound correspondences and Proto-Edoid reconstructions. We draw strictly from nouns, as verbs do not contrast for lexical tone.

Based on this database, we systematically compared tone patterns across the four proposed branches of Edoid based on the aforementioned Elugbe study: Delta Edoid (DE), South-West Edoid (SW), North-Central Edoid (NC), and North-West Edoid (NW). A number of tonal correspondences within each branch were established, which we reconstructed as combinations of tone on a noun class prefix and on the stem called nominal tone melodies (e.g. NW is reconstructed with two melodies *L-L and *H-L). The melodies reconstructed in each branch were then systematically compared against each other in order to reconstruct Proto-Edoid itself. The result is eight tonal correspondence sets reconstructed with a symmetrical set of melodies, namely **L-L, **L-LH, **L-H, **L-HL, **H-L, **H-LH, **H-HL, and **H-H (two asterisks indicate Proto-Edoid, whereas one is for an intermediate reconstruction at the branch level).

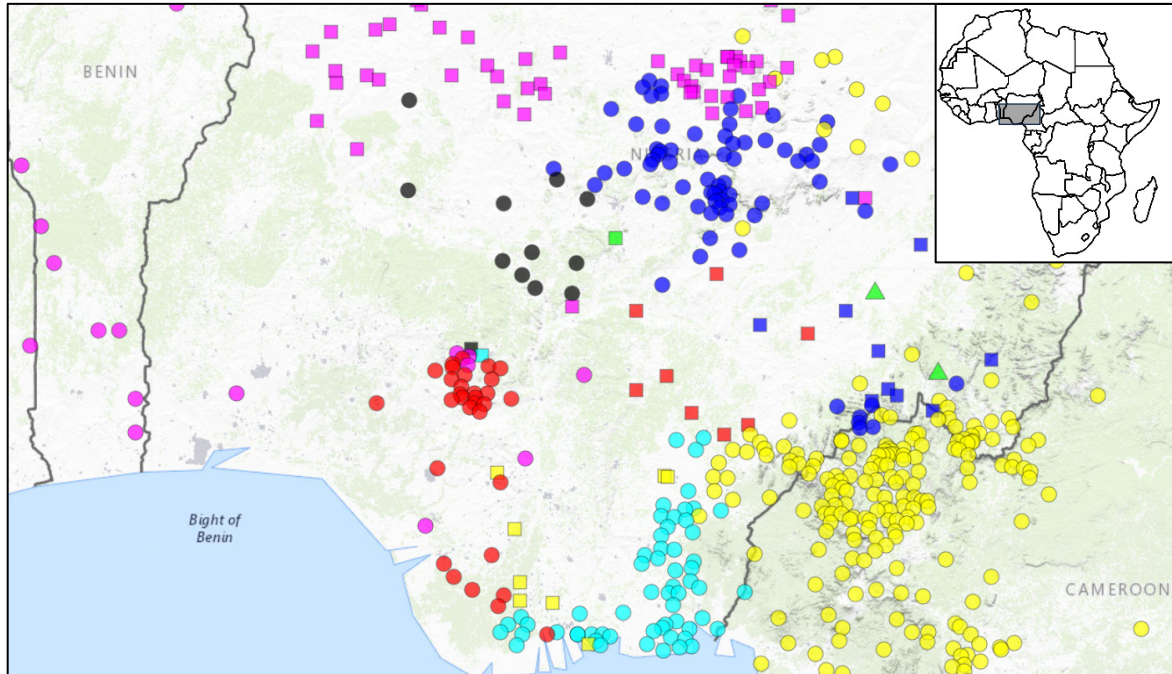
Using these PE melodies as our starting point, we track several recurring tone changes in Edoid diachrony. Two changes involve shifting of H tone, most commonly rightward shift such as the "Great Edo Tone Shift" whereby the Proto-NC *H-L melody becomes L-H in the language Edo. Leftward shift of H is also hypothesized (i.e. PE **L-H to P-NC **H-L) but this is less common, in line with the tendency for tone effects to be perseverative rather than anticipatory (Hyman 2017). Other recurrent changes include H-Obligatoriness (i.e. adding a H tone to an all-low word) and H-Lowering (i.e. H becoming L, M, or downstepped H, whether conditioned or unconditioned). These tone changes show the hallmarks of regular sound change, and thus demonstrate the success of applying the comparative method to tone melodies (though unexpected tonal outcomes for individual cognates are not uncommon and require further scrutiny).

Following this introduction (§1), we provide relevant background on Edoid (§2), followed by the data and methods of the PEdT Project (§3). We then provide the reconstructed tone melodies within the four Edoid branches (§4) and Proto-Edoid itself (§5), followed by discussion (§6) and a brief conclusion (§7).

2 The Edoid family

The Edoid family is a subgroup within Benue-Congo, itself one of the main subgroupings within the transcontinental Niger-Congo phylum (Elugbe 1989c). Edoid is spoken entirely within southern Nigeria, bordered largely by other Benue-Congo families, most prominently the Yoruboid complex to the west and the Igbooid complex to the east. The Edoid family constitutes approximately 25 to 30 languages (29 are listed on Glottolog – Hammarström *et al.* 2024). The map in (1) locates Edoid within Nigeria (the red circles ●), mapped against the other major branches of Benue-Congo (taken from Glottolog). We remain neutral as to the internal subgrouping of the major Benue-Congo branches and follow the rake-like structure of Williamson (1989:261).

- (1) Major Benue-Congo branches: Edoid ●, Yoruboid ●, Igboid ■, Delta Cross ●, Bantoid ●
 (which includes Bantu), Idomoid ■, Jukunoid ■, Nupoid ●, Platoid ●, Kainji ■



Elugbe (1973, 1989a) systematically identifies the segmental correspondences of Edoid languages, reconstructing approximately 200 Proto-Edoid words grouped into four branches based on regular sound changes and lexical innovations. The branches are geographically-labeled: North-West Edoid (NW), North-Central Edoid (NC), South-West Edoid (SW), and Delta Edoid (DE) located furthest south (in the Niger River Delta). Our project examines 21 individual Edoid languages, listed in (2) and split up based on their membership in one of the four Edoid branches. We include both the ISO code and the glottocode for each language for referential purposes. Hereafter we shall exclusively use the ISO codes, both for their brevity and their recognizability. The approximate geographic location of these 21 languages is mapped in (3), which includes several such ISO codes.

94 (2) Four branches of Edoid and the languages examined ($n=21$)²

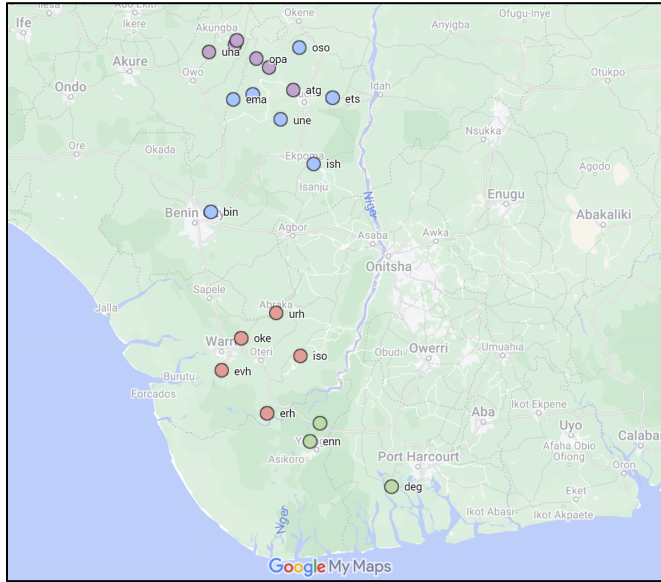
	Branch	Language	ISO	Glottocode	Coordinates	
a.	DE	Degema	[deg]	dege1246	4.73	6.81
	DE	Engenni	[enn]	enge1239	5.01	6.31
	DE	Epie-Atissa	[epi]	epie1238	5.12	6.37
b.	SW	Eruwa	[erh]	eruw1238	5.18	6.05
	SW	Isoko	[iso]	isok1239	5.53	6.25
	SW	Okpe	[oke]	okpe1250	5.64	5.89
	SW	Urhobo	[urh]	urho1239	5.79	6.10
	SW	Uvwie	[evh]	uvbi1238	5.44	5.77
c.	NC	Edo	[bin]	bini1246	6.41	5.70
	NC	Esan	[ish]	esan1238	6.70	6.33
	NC	Emai-Iuleha-Ora	[ema]	emai1241	7.09	5.84
	NC	Etsako	[ets]	yekh1238	7.10	6.45
	NC	Uneme	[une]	unem1238	6.97	6.13
	NC	Ghotuo	[aaa]	ghot1243	7.12	5.96
	NC	Ososo	[oso]	osos1238	7.41	6.25
d.	NW	North Ibie	[atg]	ivbi1241	7.15	6.21
	NW	Oloma	[olm]	olom1241	7.28	6.06
	NW	Okpamheri	[opa]	okpa1238	7.34	5.98
	NW	Uhami	[uha]	uham1238	7.38	5.69
	NW	Ehueun	[ehu]	ehue1238	7.45	5.86
	NW	Ukue	[uku]	ukue1238	7.42	5.85

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² The classification of Ososo [oso] as North-Central Edoid and North Ibie [atg] as North-West Edoid comes from Lewis (2013). The Edoid languages not surveyed in this paper are Ikhin-Arokho [ikh], Enwan [env], Ikpeshi [ikp], Igwe [igw], Sasaru [sxs], Iyayu [iya], Akuku [ayk], and Okpe [okx].

(3) Approximate coordinates of Edoid languages of this study, colored by branch

(Purple ● = NW, Blue ● = NC, Red ● = SW, Green ● = DE)



Like most languages of the area, Edoid languages have a largely analytic and head-initial syntax (e.g. verb-object order, noun-modifier order, prepositions), with limited segmental inflection. Phonologically, there are no codas and consonant clusters only involve approximants. Certain Edoid languages are known for a fortis/lenis distinction in consonants (Ladefoged 1968; Elimelech 1976; Elugbe 1989a), which cannot be straightforwardly reduced to another phonetic dimension (e.g. voicing, gemination, frication, *etc.*). Elugbe (1989a) reconstructs a full set of fortis/lenis contrasts, though this system has since collapsed in most descendent Edoid languages. In Proto-Edoid and in descendent languages which retain the fortis/lenis contrast, the lenis version is transcribed with an <h> (i.e. fortis *íkphò* ‘clothes’ vs. lenis *íkphò* ‘erosion’ in Etsako – Elimelech 1976:8). Moreover, many of the languages have active Advanced Tongue Root (ATR) harmony in the vowel system (i.e. [+ATR] *i e ə o u* vs. [-ATR] *ɪ ɛ ʌ ɔ ʊ*) also reconstructed for Proto-Edoid, though in the central core this has broken down considerably (Rolle, Lionnet, & Faytak 2020).

111 All Edoid languages are robustly tonal, mostly with a high tone (H) versus low tone (L)
112 distinction, but a few languages show evidence for a mid tone (M) (e.g. Ghotuo – Elugbe 1973,
113 1989b). Depending on the language, downstep may occur but typically only affects high tones (i.e.
114 ⁺H). The difference between M and ⁺H depends on their phonological behavior, and the two
115 transcriptions may be found even for the same language across sources (e.g. Urhobo). Contour
116 tones on syllables occur but are not particularly frequent.

117 In the entire Edoid family, only nouns have lexical tone contrasts whereas tone on verbs is
118 strictly grammatical. For example, minimal pairs in Esan exist for nouns (e.g. *áwà* ‘dog’ vs. *àwà*
119 ‘forbidden thing’) but none exist for verbs, where tone differences strictly express inflection (e.g.
120 *ghe* ‘look at’ in *Àkhèrè ghé ògó* ‘Akhere looked at a bottle’ vs. *Àkhèrè ghè ògó* ‘Akhere is looking
121 at a bottle’).

122 Edoid nouns canonically consist of a noun class prefix and a nominal stem (most often one to
123 two syllables). Across Edoid, the class prefix is almost always a single vowel though in a few
124 languages some class prefixes bear an onset. They function primarily to indicate number, with
125 class prefixes forming singular/plural pairs, as in (4). Twelve such pairings of class markers are
126 reconstructed for Proto-Edoid (Elugbe 1989a:123-124). In some languages this system is still
127 largely intact (e.g. in Oloma), while in others it is broken down with only a few nouns showing
128 any morphological alternation (e.g. in Edo).

129 (4) Edoid – Noun class prefixes forming singular/plural pairs

130 a. Oloma – System intact (Elugbe & Schubert 1976:74)

131 Singular *lé-nwè* ‘breast’ *ghó-bì* ‘leaf’

132 Plural *í-nwè* ‘breasts’ *é-bì* ‘leaves’

133 b. Edo – System broken down (Agheyisi 1990:39,87)

134 Singular *ò-kpiá* ‘man’ Cf. *è-bé* ‘leaf, leaves’

135 Plural *ì-kpiá* ‘men’

136 The noun class system plays a role in the tonal analysis to come, and we shall therefore
137 demarcate the prefix both in transcribing the word and tone pattern (e.g. for *ò-kpiá* above, the tone
138 pattern is rendered L-LH). We make this demarcation even in those words where there is no longer
139 a regular alternation (e.g. *è-bé* above). Interestingly, even in languages where the noun class
140 system is all but obsolete, there remains a requirement for all nouns to begin with a vowel, reflected
141 in recent loanwords, e.g. in Edo a prothetic *e-* is added to *è-gómù* ‘gum’, *è-fàdá* ‘Catholic
142 priest(s)’, *è-lédì* ‘lead pencil(s)’, *etc.* (Agheyisi 1986).

143 3 Data and methods

144 The PEdT Project employs the traditional “comparative method” but applied to tone patterns,
145 whereby daughter languages are systematically compared in order to reconstruct the ancestral state
146 of the mother language, ultimately Proto-Edoid (see Dockum 2019 for the comparative method
147 and its application to tone). All currently available data on Edoid was consulted for this paper.
148 Fieldwork on various Edoid languages by Ben O. Elugbe was the starting point (Elugbe 1973,
149 1977, 1986, 1989a, 1989b, 1989c; Elugbe & Schubert 1976). This seminal work was
150 complemented by an exhaustive survey of the Edoid literature, where lexical data was collected if
151 there appeared to be consistent tonal transcription (Thomas & Williamson 1967; Ladefoged 1968;

Mafeni 1969; Welmers 1969; Elimelech 1976; Thomas 1978; Iweh 1983; Aigbe 1985; Eigbedion 1985; Agheyisi 1986; Blanc 1986; Donwa-Ifode 1986, 1989; Ejele 1986; Masagbor 1989; Klomp 1993; Emuekpere-Masagbor 1997; Aziza 1997, 2003, 2008; Osiruemu 2005; Ukere 2005; Schaefer & Egbokhare 2007; Kari 2008; Olaoluwa 2015; Legbeti 2022). When possible, these data were supplemented by personal fieldnotes (specifically on Degema, Esan, Urhobo, and Okpe).

In total, the PEdT database consists of 436 cognates based on 3,038 individual data points, using only lexical items which are cognate in at least two Edoid languages. Each cognate is provided a unique code for sorting and referential purposes, as well as a cognate name, a mnemonic based on characteristic segments across languages for this cognate (note that this is not a formal segmental reconstruction), its semantic domain, the number of languages with this cognate, and its meaning in daughter languages. A sample cognate is in (5) for SMOKE_1.

(5) Sample cognate entry from the PEdT database

PE_code	PE_cognate	mnemonic	domain	languages	meaning
PE_549	SMOKE_1	UBHURI	substance	8	smoke; ashes [oke]

Following each cognate, descendent forms are provided from the relevant languages. We include the original data point (including its source, in abbreviated form – see the supplemental materials), as well as a normalized segmental and tonal rendition of the cognate, as in (6). In normalizing tone, we transcribe the isolation pronunciation (usually as provided by sources), which may or may not reflect additional tonal alternations on top of the lexical tones (we return to this issue when we discuss the surface vs. underlying tone in Degema – §4.1). When sources of the same language employ different tonal conventions or analyses, we choose one (e.g. for Urhobo in §4.2, we render all mid tones as “M” and not with downstepp “⁺H”).

173 (6) Sample cognate – SMOKE_1

	branch	language	iso	data_point	segments	tone	source
a.	DE	Engenni	[enn]	<i>uḅuru éṭaṭi</i>	<i>u-buru</i>	L-LL	{tho67}
		Epie-Atissa	[epi]	<i>uḅuru táān</i>	<i>u-buru</i>	L-LL	{tho67}
b.	SW	Isoko	[iso]	<i>ívbirrì</i>	<i>i-viri</i>	H-LL	{elu77}
		Okpe	[oke]	<i>ìirĩ</i>	<i>i-irĩ</i>	H-LL	{don89}
		Urhobo	[urh]	<i>ívìrì</i>	<i>i-vìrì</i>	H-LL	{azi97}
		Uvwie	[evh]	<i>ívìrì</i>	<i>i-vìrì</i>	H-LL	{don89}
c.	NC	Ososo	[oso]	<i>èwɔ̃</i>	<i>ε-wɔ</i>	L-L	{ola15}
d.	NW	North Ibie	[atg]	<i>èwòlì</i>	<i>e-wòli</i>	L-LL	{mas89}

174

175 Cognacy was based on established consonantal correspondences in the nominal stem, as
 176 established previously by Elugbe (1989a) (see the supplemental materials for these
 177 correspondences in full). For each of the 436 cognates, we provide a confidence rating of high,
 178 medium, or low. Those words which complied with previously established consonantal
 179 correspondences were designated with high confidence ($n=190$). A sample of such cognates is in
 180 (7). In the first two – cognates for WATER and ELEPHANT – all languages shared the same
 181 general shape. Here, the stem-initial consonant is identical and in red (and the proposed Proto-
 182 Edoid form from Elugbe 1989a is also provided at the top, indicated with two **). Other examples
 183 in (7) include cognates where there is a systematic divergence from the proto-segment in some
 184 language, such as in MARKET with ****k** and CHIN with ****gbh** (recall that *h* stands for lenis), but
 185 which Elugbe identified as regular. The bulk of changes involves the development of lenis
 186 consonants e.g. ****th** becomes DE *t* but SW *ɾ ~ r*.

	WATER (PE: **A- <i>m</i> iN)	ELEPHANT (PE: **E- <i>n</i> i)	MARKET (PE: **A- <i>k</i> i)	CHIN (PE: **A- <i>gb</i> hamhi)	PALM_1 (PE: **U- <i>d</i> i)	GREY_HAIR (PE: **ghU- <i>da</i> N)	TONGUE (PE: **U- <i>d</i> hamhi)	TREE (PE: **U- <i>th</i> aNi)	EAR (PE: **ghU- <i>ch</i> ɔGi)
a. [deg]	à- <i>mín</i>	è- <i>ní</i>	è- <i>kí</i>	ù- <i>ŋgbá</i>	ù- <i>dí</i>	ù- <i>dáŋ</i>	ù- <i>dém</i>	ó- ⁺ <i>tán</i>	ó- ⁺ <i>só</i>
[enn]	à- <i>mìn</i> i -	è- <i>kì</i>	à- <i>gbà</i>	ù- <i>dí</i>	è- <i>dàà</i>	ò- <i>dém</i> ù	é- <i>tà</i> i	é- <i>sò</i> ù	
[epi]	à- <i>mìn</i> i -	è- <i>kì</i>	à- <i>gbǎ</i>	ì- <i>dí</i>	ì- <i>dàn</i>	ì- <i>dém</i> ù	ì- <i>táān</i>	ì- <i>só</i> õ	
b. [erh]	à- <i>mí</i>	-	è- <i>kí</i>	à- <i>gbà</i>	è- <i>dí</i>	-	ù- <i>rɔvú</i>	ú- <i>rǎ</i> ǣ	-
[iso]	à- <i>mì</i>	è- <i>nì</i>	è- <i>kì</i>	à- <i>gbà</i>	ú- <i>dí</i>	-	è- <i>rɔ̀</i> ù	ú- <i>ré</i>	ú- <i>zɔ</i>
[oke]	à- <i>mì</i>	è- <i>ní</i>	é- <i>jì</i>	é- <i>gbǎ</i>	ù- <i>dì</i>	é- <i>rà</i>	ò- <i>rém</i> ù	ò- <i>rǎ</i>	ò- <i>rɔ</i>
[urh]	à- <i>mè</i>	è- <i>nì</i>	è- <i>kì</i>	é- <i>gbǎ</i>	ù- <i>dì</i>	è- <i>fà</i>	è- <i>ɬèβè</i>	ú- <i>ré</i>	ó- <i>rɔ</i>
[evh]	à- <i>mì</i>	è- <i>nì</i>	è- <i>kì</i>	à- <i>gbàvò</i>	-	-	-	ù- <i>rá</i>	è- <i>só</i>
c. [bin]	à- <i>mè</i>	è- <i>ní</i>	è- <i>kì</i>	à- <i>gbàvǔ</i> ǣ	ù- <i>dí</i>	è- <i>dé</i>	á- <i>rǎvǔ</i> ǣ	è- <i>rǎ</i>	è- <i>hó</i>
[esa]	à- <i>mè</i>	é- <i>nì</i>	è- <i>kì</i>	à- <i>gbǎ</i>	ú- <i>dí</i>	-	ó- <i>là</i> mè	é- <i>rǎ</i>	é- <i>hò</i>
[ema]	à- <i>mè</i>	í- <i>nì</i>	è- <i>kì</i>	à- <i>gbǎ</i>	é- <i>dí</i>	é- <i>dè</i>	ó- <i>è</i> mì	ó- <i>rǎ</i>	é- <i>hò</i>
[ets]	à- <i>mè</i>	í- <i>nì</i>	à- <i>kì</i>	à- <i>gbà</i>	ú- <i>dì</i>	é- <i>dè</i>	ó- <i>lè</i> mhi	ó- <i>rà</i>	é- <i>sò</i>
[une]	à- <i>mè</i>	í- <i>nì</i>	à- <i>kì</i>	à- <i>gbà</i>	ú- <i>dì</i>	é- <i>dè</i>	é- <i>řè</i> mhi	ó- <i>θà</i>	é- <i>sò</i>
[aaa]	à- <i>mè</i>	í- <i>ní</i>	ghò- <i>kì</i>	à- <i>gbà</i>	ghí- <i>lì</i>	ghò- <i>dè</i>	ghí- <i>lè</i>	ō- <i>tà</i>	ghō- <i>hò</i> yì
[oso]	à- <i>mè</i>	-	ò- <i>xì</i>	[í-zù]à- <i>gbà</i>	-	-	írè- <i>rè</i>	ó- <i>thè</i>	ó- <i>sò</i>
d. [atg]	à- <i>mè</i>	-	-	è- <i>kpà</i> mì	-	é- <i>dè</i>	-	ó- <i>rè</i>	é- <i>sò</i>
[olm]	à- <i>mè</i>	é- <i>nì</i>	ghè- <i>kì</i>	à- <i>gbà</i>	ghú- <i>dì</i>	ghó- <i>zè</i>	ó- <i>lè</i> mhi	ó- <i>shà</i>	ghó- <i>žò</i>
[opa]	à- <i>mè</i>	-	wè- <i>kì</i>	à- <i>gbà</i>	-	-	ó- <i>rè</i>	ú- <i>fà</i>	wó- <i>žò</i>
[uha]	à- <i>mè</i>	è- <i>nì</i>	è- <i>cì</i>	à- <i>gbà</i> mì	ì- <i>dì</i>	-	è- <i>dò</i> mì	ó- <i>fà</i>	é- <i>sò</i>
[ehu]	à- <i>mè</i>	-	è- <i>hì</i>	à- <i>wǎ</i>	-	-	-	è- <i>fā</i>	ē- <i>ɔ̀</i>
[uku]	à- <i>mè</i>	-	è- <i>kì</i>	à- <i>gbǎ</i>	í- <i>dì</i>	-	é- <i>dà</i> mì	ò- <i>fā</i>	é- <i>ɔ̀</i>

188

189 There were also cognates rated with medium confidence (n=171). These words deviated from
190 the expected segment correspondence typically with respect to the stem-initial consonant, a sample
191 of which is provided in (8). For example, ANTELOPE_2 has a correspondence between *j* in DE
192 languages and *z* in the other branches which is not otherwise established in Elugbe (1989a)
193 (perhaps ***j* ~ ***j* variation in Proto-Edoid for this word). We assume cognacy if there is both a
194 shared meaning, and there are additional non-trivial phonological properties in common, e.g. in

195 terms of sharing a major consonantal feature (e.g. place of articulation), containing the same
 196 number of syllables, sharing vowel quality, *etc.*

197 (8) Medium confidence – Non-established correspondence but plausible cognacy ($n=171$)

	ANTELOPE_2 (j↔z)	MUSHROOM (s↔t)	CALABASH_4 (gb↔b↔β)	FOOT_LEG_2 (β/v/ʋ↔w↔ Ø↔mh)	FOOD (m↔v↔b↔n)	COUGH_N (j↔w↔hw↔ ηm↔h)	SNAIL (s↔r↔r↔ θ↔l↔th)	BLOOD_2 (z↔gj↔dʒ↔ d↔dz↔r)
a. [deg]	ú- ⁺ jów	ù-sùdú	-	à-βí	-	-	-	ì-zàlà
[enn]	ú-jòu	-	à-gbìnà	à-vì	-	-	-	-
[epi]	-	-	è-gbèlè	à-vì	-	ù-jílō	-	ì-zàrà
b. [erh]	-	-	-	à-vò	-	-	-	à-zí
[iso]	-	-	ò-bèlè	ù-vò	-	-	ù-sèkpè	à-zè
[oke]	-	-	-	ò-wò	è-máré	-	-	-
[urh]	ú-zò	ì-tú	-	ò-wò	è-màré	ó-wòrè	ú-sèkpè	-
[evh]	-	-	è-bèlè	à-vò	-	-	-	-
c. [bin]	ú-zò	ù-tú	-	ò-wè	è-vàré	ó-hwé	ú-réé	è-sájjè
[esa]	ú-zò	ú-tú	-	ò-è	é-bàlè	ó- ⁺ hwé	ú- ⁺ rè	-
[ema]	ú- ⁺ zó	í-tùú	ù-bèlè	ò-è	é-màè	ó-ηmèé	ú- ⁺ rè	è-sádʒè
[ets]	-	-	ù-bènè	ò-wè	-	ó-jè	ú-rè	-
[une]	-	-	-	-	-	-	ú-θèì	-
[aaa]	-	-	ù-bèlè	-	-	-	ū-ré	ò-òdè
[oso]	-	-	ù-βèrè	ò-wè	é-nèrè	-	-	ò-zè
d. [atg]	-	-	-	ò-wè	-	-	ú-lùé	-
[olm]	-	-	-	ghô-mhìnà	-	ghó-hò	ú-thùnhámì	à-zè
[opa]	-	-	-	-	-	-	-	à-dzè
[uha]	-	-	-	-	-	-	-	è-zè
[ehu]	-	-	-	-	-	-	-	è-rè
[uku]	-	-	-	-	-	-	-	è-dè

198

199 Lastly are cognates rated with low confidence ($n=75$), exemplified in (9). This rating was
 200 given to cognates which only exist for one Edoid branch, where cross-branch comparison is not
 201 possible. Here, we can speculate as to the Proto-Edoid form but further comparison with another
 202 branch would be needed for confirmation.

203 (9) Cognates with low confidence ($n=75$)

		NAVEL_3	FOREST_3	BODY_3	HAWK_3	CHAIN	TOMORROW	SKIN_2	CORPSE_2
a.	[deg]	ù-dúm	è-dzí	-	-	-	-	-	-
	[enn]	ù-dùmù	à-zhì	-	-	-	-	-	-
	[epi]	ù-dùmù	à-zì	-	-	-	-	-	-
b.	[erh]	-	-	ó-mà	-	-	-	-	-
	[iso]	-	-	ò-mà	ò-sò	-	-	-	-
	[oke]	-	-	ò-mà	-	-	-	-	-
	[urh]	-	-	ò-mà	ò-sò	-	-	-	-
	[evh]	-	-	ò-mà	ó-sù	-	-	-	-
c.	[bin]	-	-	-	-	é-yà	á-xwè	-	-
	[esa]	-	-	-	-	ì-yà	á-xò	-	-
	[ema]	-	-	-	-	ì-yà	á-xò	-	-
	[ets]	-	-	-	-	à-yà	á-xwè	-	-
	[une]	-	-	-	-	ì-gà	á-xò	-	-
	[aaa]	-	-	-	-	ì-yà	-	-	-
	[oso]	-	-	-	-	-	à-kò	-	-
d.	[atg]	-	-	-	-	-	-	-	-
	[olm]	-	-	-	-	-	-	ghé-shì	-
	[opa]	-	-	-	-	-	-	wé-sì	ó-khùù
	[uha]	-	-	-	-	-	-	ì-sì	ò-gù
	[ehu]	-	-	-	-	-	-	ì-rì	ò-gù
	[uku]	-	-	-	-	-	-	ì-rì	ó-gù

204

205 All three of these confidence grades – high, medium, and low – are included in the PEdT
 206 database (and can be included or excluded at the discretion of the user). At the same time, there
 207 were numerous words outside of this scale which were explicitly excluded from the database. The
 208 majority of these are known or suspected loanwords, e.g. the word for ‘sea, ocean’ in Edo and
 209 Emai is òkũ which is nearly identical to the Yoruba [yɔɾ] form òkũ. Other excluded words are
 210 *wanderwörter* which are found widely in Southern Nigeria, e.g. the word for ‘lion’ which is ò-

dúmà in Engenni, *ò-dùmá* in Edo and Emai, and *ò-dùmhâ* in Etsako (all Edoid languages), but also *òdúm* in Igbo [i b o] and *ìdùm* in Kalabari [i j n] (non-Edoid languages).

4 Reconstructed nominal tone melodies for each Edoid branch

In this section, we present our reconstructions of the four individual branches, namely Delta Edoid (DE), South-West Edoid (SW), North-Central Edoid (NC), and North-West Edoid (NW). As stated, Elugbe (1989a) establishes this four-way split based on segmental reconstruction, but tone played no role in this proposal.

Though the PEdT database consists of 436 cognates, the four branches vary as to how many reflexes of these cognates are present. As shown in Table (10), this ranges from over 86% in NC but half that many in NW at just under 43%. Given that the NC branch as a whole is probably the best documented while NW is probably the worst, their high and low cognate percentages may simply be due to their varying documentation levels.

(10) Number of cognates found in each Edoid branch compared to total ($n=436$)

Branch	Cognates	Percentage of total
Delta (DE)	227	52.1%
South-West (SW)	297	68.1%
North-Central (NC)	377	86.5%
North-West (NW)	186	42.7%

As a preview, (11) summarizes the tonal correspondence sets within each branch: four in DE, five in SW, five in NC, and two in NW. For each correspondence set, we provide our proposed tonal reconstruction. These are of the shape $*T_1-T_2(T_3)$, where T_1 indicates the tone on the prefix and T_2 (and T_3 , when present) indicate the tone of the stem. We refer to these reconstructions as “nominal tone melodies”. The number of cognates belonging to a tone melody is highly skewed: $*L-L$ is by far the most common in DE, $*L-L$ and $*H-L$ are both roughly equally dominant in SW and NC, and $*H-L$ is most common in NW.

(11) Summary of tonal correspondence sets and reconstructed nominal tone melodies

Branch	Correspondence set	Reconstruction	<i>n</i> =
a. Delta	DE-01	*L-L	125
	DE-02	*H-H	39
	DE-03	*H-L	19
	DE-04	*L-H	19
b. South-West	SW-01	*L-L	91
	SW-02	*L-LH	38
	SW-03	*L-H	48
	SW-04	*H-L	78
	SW-05	*H-H	21
c. North-Central	NC-01	*L-L	121
	NC-02	*H-L	118
	NC-03	*H-LH	53
	NC-04	*H-H	27
	NC-05	*L-H	27
d. North-West	NW-01	*L-L	56
	NW-02	*H-L	103

Importantly, the nominal tone melodies constitute word-level patterns and are not intended to be reconstructions of individual tonemes. A reconstructed tone *H-L indicates that the prefix bore H tone and the stem bore a L tone which was distributed across the entire stem. The synchronic tone systems of Edoid languages tend to operate in terms of tone melodies in this manner, where the majority of nouns fall into a small number of patterns. For example, in Degema nearly 80% of all nouns fall into three word-level tone patterns (see Rolle & Kari 2022:259 for details).

4.1 Delta Edoid (DE)

Delta Edoid (DE) contains three languages: Degema [deg], Epie-Atissa [epi], and Engenni [enn]. Degema is the best described, including a grammar (Kari 2004), a dictionary (Kari 2008), and a worked-out synchronic analysis of the tone system (Rolle & Kari 2022). We can establish 227 cognates showing reflexes in the DE branch, just over half the total number of Edoid cognates. We can establish four primary tonal correspondence sets. The first and largest correspondence set is in (12), which we reconstruct as a Proto-DE nominal tone melody *L-L, and label it neutrally

as DE-01 ($n=125$). This correspondence shows an all-low pattern in Engenni and Epie-Atissa, but a low plus final high pattern in Degema.

(12) Examples of DE-01, the *L-L melody ($n=125$)

Cognate	deg	enn	epi	Proto-Edoid
a. DOCTOR (PE_187)	ǎ- <i>bú</i> L-H	ǎ- <i>bù</i> L-L	ǎ- <i>bù</i> L-L	** <i>O-bu</i>
b. GREY_HAIR (PE_281)	ǎ- <i>dáŋ</i> L-H	ǎ- <i>dàà</i> L-L	ǎ- <i>dàn</i> L-LL	** <i>ghU-daN</i>
c. ANIMAL_MEAT (PE_010)	ǎ- <i>nám</i> L-H	ǎ- <i>nàmù</i> L-LL	ǎ- <i>nàmù</i> L-LL	** <i>E-nhamɿ</i>
d. ASH (PE_023)	ǎ- <i>màfú</i> L-LH	ǎ- <i>mùfù</i> [<i>mù</i>] L-LL[L]	ǎ- <i>wòfù</i> L-LL	** <i>A-mhuNəfu</i>

These tonal transcriptions represent the surface pronunciation in isolation. Crucially, Rolle and Kari (2022) show that this final H tone in Degema is actually not present in most contexts, and consequently they analyze this pattern as /L-L/ at a more abstract phonological level. Thus, the underlying forms in (12) would be akin to /ǎ-*bù*/ which surface as [ǎ-*bú*]. This renders this tone melody as a whole as stable, not deviating from the proposed Proto-DE *L-L phonological representation in any daughter language.

Individual cognates may deviate from our expected correspondence, requiring us to posit either a separate correspondence set, or to split the existing one into multiple subsets. In most cases, we follow the latter strategy. For example, in (13) cognates which show the expected all-low pattern are part of subset DE-01A (by far the most numerous – $n=69$), while a small minority of cognates for which the Engenni form has a high tone somewhere are labelled DE-01B ($n=3$). Likewise, where Epie-Atissa has an unexpected H somewhere in the word, cognates are labeled DE-01C, and where Degema has a final L, cognates are labeled DE-01D. We suspect these deviations are due to effects of a compound construction.

265 (13) Subsets for DE-01 *L-L

Subset	<i>n</i> =	Cognate	deg	enn	epi
DE-01A	69	DEATH	ù-wú L-H	ù-wù L-L	ù-wù L-L
DE-01B	3	HAND_ARM	ò-bó L-H	ó-bò H-L	ò-bò L-L
DE-01C	7	BIRD	è-fén L-H	à-fèni L-LL	è-féní L-HH
DE-01D	7	BLOOD_2	ì-zàlà L-LL	-	ì-zàrà L-LL
DE-01U	39	BILE	-	è-lò L-L	-

266

267 The last subset DE-01U are cognates which fit into multiple subsets due to the lack of relevant
 268 cognates in the other languages (here, “U” stands for “unspecified”). Moreover, when a language
 269 has no data point, we denote this with a dash.

270 The *L-L melody is as large as all the other classes combined, one of which is a tone melody
 271 *H-H (DE-02 – *n*=39), exemplified in (14). Here, Degema appears with a H-⁺H pattern (with
 272 downstep), Engenni as H-L, and Epie-Atissa as L-HM. As above, some reflexes have an
 273 unexpected form: an initial H on the prefix in Epie-Atissa (DE-02B), or an all-low pattern in
 274 Engenni (DE-02C) or Epie-Atissa (DE-02D).

275 (14) Examples of DE-02, the *H-H melody (*n*=39)

Subset	<i>n</i> =	Cognate	deg	enn	epi
DE-02A	11	DOG_1	á- ⁺ búwá H- ⁺ HH	á-bùà H-LL	à-búā L-HM
DE-02B	5	RIVER_1	é- ⁺ dá H- ⁺ H	é-dà H-L	é-dē H-M
DE-02C	3	WORM_1	-	à-gìò L-LL	á-gān H-M
DE-02D	2	SLAVE_2	ó- ⁺ kóní H- ⁺ HH	-	ò-kòni L-LL
DE-02U	12	BASKET_2	ú- ⁺ túwó H- ⁺ HH	ú-tùò H-LL	-
DE-02X	6				

As before, several cognates lack a form in Epie-Atissa, which prevents us from ascertaining its subset; we label these DE-02U. Lastly, those labeled DE-02X are miscellaneous patterns with other minor but unexpected deviations. By definition, these do not have a uniform pattern across the three languages, and we therefore do not provide a sample cognate.

Rolle and Kari (2022) also establishes that Degema inserts a downstep between two high tones at the end of a phonological phrase (...HH → ...H⁺H). Given that downstep is derived, we may posit a simpler /H-H/ which surfaces as [H⁺H] in Degema when the word is phrase-final (e.g. RIVER_1 /é-dá/ → [é-⁺dá]). Two sound changes would then have taken place for the other DE languages. First, in Engenni the stem high becomes low. Second, in most words in Epie-Atissa the prefix high tone is shifted rightward one syllable, which results in the /L-HM/ forms transcribed in the sources (where “M” in Epie-Atissa and “⁺H” in Degema are equivalent). At the same time, some words do not show the prefix L- and instead exhibit the same initial H as the other languages (e.g. DE-02B). At this point, no discernible conditioning environment has been found to explain the split here.

The final two DE correspondence sets are much smaller. One is reconstructed as a *H-L melody (DE-03 – *n*=19), largely based on data from Degema. In this set, Degema has a H-HL (or H-L) form which corresponds to an initial H or L in Engenni, but an initial L (if a prefix is present) in Epie-Atissa. However, for most of these cognates we only find a data point for Degema (DE-03U).

295 (15) Examples of DE-03, the *H-L melody ($n=19$)

Subset	$n=$	Cognate	deg	enn	epi
DE-03A	5	NOSE	<i>í-súβèŋ</i> H-HL	<i>ú-sùèì</i> H-LLL	<i>ù-súbē</i> L-HM
DE-03B	2	TODAY	<i>í-nínà</i> H-HL	<i>ì-njá</i> L -HL	<i>ø-jínēnē</i> ø-HMM
DE-03U	12	COLD_N	<i>ó-níni</i> H-HL	-	-

296

297 The final set is reconstructed as *L-H, labeled DE-04 in (16). These have a L-H⁺H pattern in
 298 Degema (presumably from a H-tone breaking operation at some point), but a L-HL and L-HM
 299 pattern in Engenni and Epie-Atissa, respectively. Such patterns mirror the behavior of high tones
 300 in the *H-H series (DE-02) in (14) above: downstepped highs in Degema corresponds to low in
 301 Engenni and mid in Epie-Atissa.

302 (16) Examples of DE-04 *L-H ($n=19$)

Subset	$n=$	Cognate	deg	enn	epi
DE-04A	7	MORNING	<i>à-wí⁺jé</i> L-H ⁺ H	<i>à-wíè</i> L-HL	<i>è-wíē</i> L-HM
DE-04B	2	VILLAGE_1	<i>ò-wé⁺éj</i> L-H ⁺ H	<i>ò-gwè</i> L-L	-
DE-04U	5	NEEDLE_2	<i>ù-dí⁺dĩŋ^w</i> L-H ⁺ H	-	-
DE-04X	5				

303

304 Our proposal for Proto-DE is summarized in (17), which provides the four correspondence
 305 sets with their reconstructed value, their canonical tonal reflex in the descendent languages, and
 306 the number and percentage from the total ($n=227$). Engenni and Epie-Atissa show several mergers,

which is indicative of the Delta branch as a whole. We will see this especially when we examine postulated changes from Proto-Edoid to Proto-DE (§5).³

(17) Four primary tonal correspondence sets in Delta Edoid

Proto-DE	Set	deg	enn	epi	Cognates	Total ($n=227$)
*L-L	DE-01	L-H	L-L	L-L	126	55.1%
*H-H	DE-02	H- ⁺ H	H-L	L-HM/H-HM	39	17.2%
*H-L	DE-03	H-HL	H-L	L-HM	19	8.4%
*L-H	DE-04	L-H ⁺ H	L-HL	L-HM	19	8.4%
Other					25	11.0%

In (17), there are 25 cognates which are labeled other, which do not fit into these four sets but are not in sufficient numbers to warrant a separate correspondence set. For example, the cognate for TWENTY is [ijów] in Degema, [ijèi] in Engenni, and [ijéè] in Epie-Atissa, a correspondence L-H↔H-L↔L-HL not otherwise found. As it stands, these are interpreted simply as irregularities with the hope that future research can shed light on these forms (see §6.2 for extended discussion).

4.2 South-West Edoid (SW)

Moving north, the next branch is South-West Edoid (SW) containing five languages: Eruwa [erh], Isoko [iso], Okpe [oke]⁴, Urhobo [urh], and Uvwie [evh], of which Urhobo is the largest and best described. There were 297 cognates with reflexes in this branch, roughly two thirds of the total number of cognates ($n=436$).

³ We note that in DE, it may be the case that *H-H and *H-L can be collapsed into one proto-series, which would entail a split in Degema rather than a merger in the latter two DE languages. In Degema, trisyllable words show a near complementary distribution: [H-HL] words exist but [H-LL] are rare, and [H-⁺HH] is allowed but [H-H⁺H] is not. This may speak to an earlier process where of *H-X (which could be either *H-H or *H-L, as there would be no contrast) become H⁺H or HL in Degema depending on syllable count.

⁴ Note that Okpe with ISO code [oke] is different from the identically named Edoid language Okpe with ISO code [okx]. Okpe-[oke] belongs to the South-West branch and is surveyed here, while Okpe-[okx] belongs to the North-West branch and is not surveyed in this study (see Lewis 2013 for some data on Okpe-[okx]).

The first correspondence class is reconstructed as *L-L (SW-01), in (18). For this melody, the majority of cognates in Eruwa show a L-H pattern, for Okpe a H-L pattern, and for the other three SW languages L-L (SW-01A). In a minority, Eruwa has a L-L pattern and Okpe shows L-L as well (SW-01B and C). However, we must note that for the majority of cognates we have data only from Urhobo (SW-01U below), which is ambiguous on its own as to this patterning.

(18) Examples of SW-01 *L-L ($n=91$)

Subset	$n=$	Cognate	erh	iso	oke	urh	evh
SW-01A	19	TOOTH	à-kɔ́ L-H	à-kɔ̀ L-L	á-wɔ̀ H-L	à-kɔ̀ L-L	à-kɔ̀ L-L
SW-01B	2	CHARM_DRUG_3	ù-mù L-L	ù-mù L-L	ú-ũũ H-L	ù-xùvù L-LL	ù-mù L-L
SW-01C	10	FOOT_LEG_2	à-vɔ́ L-L	ò-wɔ́ L-L	ò-wɔ́ L-L	ò-wɔ́ L-L	à-vɔ́ L-L
SW-01D	8	FEAR_2	-	ò-žɔ́ L-L	-	ò-ɟɔ́ L-L	ò-ɟɔ́ L-L
SW-01U	48	BALD_N	-	-	-	à-kpǎ L-L	-
SW-01X	4						

Despite the presence of H tones in Eruwa and Okpe, we reconstruct this as *L-L in Proto-SW for three reasons. First, this is the pattern actually seen in a minority of cognates (SW-01C above). Second, there is variation across the Eruwa and Okpe sources between a [L-L] form and one with a high tone, for example Eruwa BAG_1 is [è-kpá] ~ [è-kpà], EYE is [à-rú] ~ [à-rò], LOUSE_1 is [ù-zú] ~ [ù-zù], and AXE_1 is [è-zúé] ~ [è-zùè], depending on the source. Third, we see several places in Edoid where all-low patterns add an epenthetic high tone as a synchronic rule in the language (seen already in Degema in §4.1, and to be seen in Oloma in §4.4). We therefore posit a rule which added a final H to all-low patterns in Eruwa and an initial H in Okpe in the same context.

336 It is unclear if this is an active alternation in these languages, or what conditions its non-application
 337 in a minority of cases (i.e. SW-01C).⁵

338 There are two other sizable correspondence sets which involve an initial L tone on the prefix.
 339 One is reconstructed as *L-LH (SW-02), in (19), where Eruwa and Okpe have a final high tone
 340 while Isoko, Urhobo, and Uvwie have all low tones (with some minor subsets, as indicated). This
 341 distribution mimics that of *L-L (SW-01) in (18) where it was also Eruwa and Okpe which exhibit
 342 a high tone somewhere. The motivation for analyzing SW-02 as *L-LH (but SW-01 as *L-L) is
 343 that we shall see that SW-02 class corresponds to cognates with a high tone in NC and NW Edoid
 344 branches. In contrast, SW-01 primarily corresponds to all low-toned classes in those same
 345 branches.

346 (19) Examples of SW-02 *L-LH ($n=38$)

Subset	$n=$	Cognate	erh	iso	oke	urh	evh
SW-02A	27	MOUTH	è-nú L-H	ù-nù L-L	ù-nú L-H	ù-nù L-L	è-nù L-L
SW-02B	5	TONGUE	ò-ròvú L-LH	è-ròù L-LL	ò-rémù L-HL	è-ɣèβě L-LL	-
SW-02C	2	DOCTOR	ǝ-bò L-L	ǝ-bò L-L	ǝ-bó L-H	ǝ-bò L-L	ǝ-bò L-L
SW-02X	4						

347

⁵ One potential explanation may involve the fortis/lenis distinction from Proto-Edoid (since broken down in SW). For example, words belonging to SW-01A which condition the H-addition rule include Proto-Edoid TOOTH ***dhI-koN*, GREY_HAIR ***ghU-daN*, SUN ***U-vuNə*, WATER ***A-miN*, and WAR_FIGHT_2 ***dhI-khumhi*. The last of these cognates is the only one with a reconstructed lenis consonant (the voiceless velar stop ***kh*). In contrast, words of the SW-01C subset which shows all-low patterns across SW languages are more prone to have an initial lenis consonant, e.g. EYE ***dhI-dhu*. However, reconstructed fortis consonants also exist here (e.g. LIFE_1 ***A-gboN*).

348 From these data, we must posit a diachronic change which merged the *L-LH and *L-L melodies
 349 in Isoko, Urhobo, and Uvwie, where the final H was lost. In Eruwa and Okpe, this changed to L-
 350 H and therefore merged with the *L-H melody (immediately below).

351 Another melody is *L-H (SW-03), in (20), which remains unchanged across all five SW
 352 languages (with minor variations). We include in the *L-H melody cognates which have L-HL
 353 across the SW languages (SW-03B), in order to have a large enough number for comparison to
 354 other Edoid branches later (§5).

355 (20) Examples of SW-03 *L-H ($n=48$)

Subset	$n=$	Cognate	erh	iso	oke	urh	evh
SW-03A	28	GOAT_1	è-ví L-H	è-ví L-H	è-vé L-H	è-vé L-H	è-ví L-H
SW-03B	6	ELDER_4	ò-kpákò L-HL	ò-kpákò L-HL	ò-kpákò L-HL	ò-kpákò L-HL	ò-kpákò L-HL
SW-03C	3	HUNTER	-	ó-zúé H-HH	ò-rúé L-HH	ò-rúé L-HH	ò-rúé L-HL
SW-03X	11						

356
 357 Our final major correspondence sets are both constructed with an initial H on the prefix: a *H-
 358 L melody (SW-04) and a *H-H melody (SW-05). Compared to the L-initial reconstructed forms
 359 above, the H-initial forms show a higher degree of variation in the descendent forms, making it
 360 much more difficult to ascertain what should be treated as a subset from what should be treated as
 361 a new correspondence set. First consider the *H-L melody in (21), which has a H-L pattern across
 362 all SW languages (SW-04A). However, a number of cognates in Urhobo (SW-04B) instead show
 363 a H-M correspondent containing a mid tone (which might alternatively be transcribed as H-[↑]H
 364 with a downstep). At this point, we classify these as a single melody *H-L (as opposed to two
 365 classes *H-L and *H-M which merged in all SW but Urhobo), because neither the /H-L/ nor the
 366 /H-M/ reflexes in Urhobo cleanly correspond to distinct sets in other Edoid branches. Moreover,

in other SW languages /H-L/ patterns can be optionally realized as [H-M] such as in Isoko (Elugbe 1977). Even in Urhobo, the /H-M/ words are realized as [H-L] when they are in non-phrase final position (Welmers 1969). At this point, we simply note this discrepancy in Urhobo and await further study of a possible conditioning factor.

(21) Examples of SW-04 *H-L ($n=78$)

Subset	$n=$	Cognate	erh	iso	oke	urh	evh
SW-04A	33	SMOKE_1	-	<i>í-vìrrì</i> H-LL	<i>í-ìrì</i> H-LL	<i>í-vìlì</i> H-LL	<i>í-vìrì</i> H-LL
SW-04B	17	GROUND	<i>ú-ròrì</i> H-LL	<i>ú-tò</i> H-L	<i>ó-tòrì</i> H-LL	<i>ó-t̃</i> H-M	<i>ó-r̃ò</i> H-L
SW-04C	4	BONE_1	<i>ù-βè</i> L-L	<i>ú-ηwè</i> H-L	<i>ú-vè</i> H-L	<i>í-wē</i> H-M	<i>ú-wè</i> H-L
SW-04D	5	YESTERDAY	<i>ó-dè</i> H-L	<i>ò-dè</i> L-L	<i>ó-dè</i> H-L	<i>ó-dē</i> H-M	-
SW-04E	6	ROPE_2	<i>ú-fì</i> H-L	<i>ú-fì</i> H-L	<i>ù-fì</i> L-H	<i>ú-fì</i> H-M	<i>ú-fì</i> H-L
SW-04U	7	URINE_2	<i>á-kà</i> H-L	-	<i>é-kà</i> H-L	-	<i>á-kà</i> H-L
SW-04X	6						

The last set is reconstructed as *H-H, in (22), where Isoko and Urhobo retain the /H-H/ pattern while the other three SW languages have condensed the high span to a single position, either to the prefix (in Eruwa) or the stem (Okpe and Uvwie). However, like we saw with the *H-L set in (21), there are frequent deviations from the expected reflexes, e.g. lowering in Isoko (SW-05B).

(22) Examples of SW-05 *H-H ($n=21$)

Subset	$n=$	Cognate	erh	iso	oke	urh	evh
SW-05A	6	TREE	<i>ú-ràrè</i> H-LL	<i>ú-ré</i> H-H	<i>ò-r̃á</i> L-H	<i>ú-ré</i> H-H	<i>ò-rá</i> L-H
SW-05B	3	HEART	<i>[ú-ví]-ú-dù</i> [H-H]-H-L	<i>ú-dù</i> H-L	-	<i>ú-dú</i> H-H	-
SW-05U	5	TRAP	-	-	-	<i>ú-wé</i> H-H	-
SW-05X	7						

We summarize the five SW correspondence sets in (23).

(23) Five major correspondence sets in South-West Edoid

Proto-SW	Set	erh	iso	oke	urh	evh	Cognates	Total ($n=297$)
*L-L	SW-01	L-H/L-L	L-L	H-L/L-L	L-L	L-L	91	30.6%
*L-LH	SW-02	L-H	L-L	L-H	L-L	L-L	38	12.8%
*L-H	SW-03	L-H	L-H	L-H	L-H	L-H	48	16.2%
*H-L	SW-04	H-L	H-L	H-L	H-L/H-M	H-L	78	26.3%
*H-H	SW-05	H-L	H-H	L-H	H-H	L-H	21	7.0%
Other							21	7.0%

4.3 North-Central Edoid (NC)

Moving further north, the next branch is North-Central Edoid (NC) which includes the languages Edo [bin], Esan [ish], the Emai lect of the Emai-Ora-Iuleha cluster [ema], Etsako [ets], Uneme [une], Ghotuo [aaa], and Ososo [oso]. Notably, the NC branch contains the Edo language (also known as Bini), the language of the Benin Empire which dominated this region of Nigeria from the 15th to 19th centuries (Egharevba 1968 [1934]; Ryder 1969).

Two correspondence sets together account for over 50% of NC cognates: one reconstructed as *L-L (NC-01, $n=121$) and one reconstructed as *H-L (NC-02, $n=119$). The proto-NC *L-L melody is exemplified in (24), where all seven NC languages retain the all-L pattern (NC-01A). For each NC language, however, a high tone may sporadically appear, typically on the prefix (NC-01B to NC-01H). The most notable of these is Edo (subset NC-01B), which have the largest number of words with an initial H tone ($n=14$).

394 (24) NC-01 *L-L (*n*=121)

Subset	<i>n</i> =	Cognate	bin	ish	ema	ets	une	aaa	oso
NC-01A	59	MARKET	è-kì L-L	è-kì L-L	è-kǐ L-L	à-kì L-L	à-kì L-L	γò-kì L-L	ò-xì L-L
NC-01B	14	CHAIN	é-yǎ H-L	ì-yǎ L-L	ì-yǎ L-L	à-yà L-L	ì-gà L-L	ì-yà L-L	-
NC-01C	2	FRIEND_1	ò-tù L-L	ǎ-siò H-L	-	-	-	ǎ-sè L-L	ǎ-tfi[àrì] L-L[LL]
NC-01D	4	AGE_GRADE_1	ò-tù L-L	ò-tù L-L	ò-tú L-H	ò-tù L-L	ò-tù L-L	-	-
NC-01E	1	FOREST_2	-	-	ò-gò L-L	ǎ-gwà H-L	-	-	è-gwà L-L
NC-01F	1	PROFIT	è-rè L-L	-	è-è L-L	è-lèlè L-LL	ǎ-lé F-H	è-lèlì L-LL	-
NC-01G	2	JUJU_MAGIC	è-bò L-L	-	-	-	-	ǎ-wò M-L	[iz]ò-bò [L]L-L
NC-01H	4	SUN	ò-vǎ L-L	ò-vwǎ L-L	ò-vǎ L-L	ò-vǎ L-LL	ò-vǎrì L-LL	ò-vǎ L-L	ǎ-vǎ H-L
NC-01U	20	BOUNDARY_3	-	-	ù-hì L-L	-	-	-	-
NC-01X	14								

395

396 The second largest set is the *H-L melody, in (25). While the reflex of this is H-L in most NC
 397 languages, in Ghotuo [aaa] it is M-L (which contrasts with high tone), and in Edo it appears to
 398 have shifted one syllable rightwards (i.e. *H-L > L-H). This reconstruction entails a “Great Edo
 399 Tone Shift” whereby a formerly initial high tone on the prefix moved onto the stem.⁶

⁶ The “Great Edo Tone Shift” is an allusion to similarly named diachronic tone changes: the “Great Igbo Tone Shift” (Hyman 1974) and the “Great Ngamo Tone Shift” (Schuh 2017:135).

400 (25) Reflexes of NC-02 *H-L ($n=119$)

Subset	$n=$	Cognate	bin	ish	ema	ets	une	aaa	oso
NC-02A	47	DOG_1	à-wá L-H	á-wà H-L	á-wà H-L	á-ɣwà H-L	á-wà H-L	ɣā-wà M-L	á-wà H-L
		HEAD	ù-húvũ L-HL	ú-hǝ̃β̃ H-LL	ú-hũmì H-LL	ú-sò H-L	ú-fòmhi H-LL	ũ-sò M-L	ú-kwè H-L
NC-02B	16	CHICKEN	ḡ-xóxò L-HL	ḡ-xóxò H-HL	ḡ-ḡxò H-HL	ḡ-xóxò H-HL	ḡ-xóxò H-HL	ḡ-hò M-L	ḡ-xòxò H-LL
NC-02C	11	CLOTH_1	ù-kpḡ L-L	ú-kpḡ H-L	ú-kpũ H-L	ú-kpò H-L	ú-kpò H-L	ũ-kpò M-L	-
NC-02D	2	NECK_THROAT_2	è-hóò L-HL	à-xòlè L-LL	-	-	-	é-hàlì H-LL	-
NC-02E	2	BLOOD_1	è-rǎḡ L-HL	á-rǎḡ H-LL	è-rèè L-LL	á-rǎi H-LL	í-θàì H-LL	-	-
...									
NC-02U	9	GIFT	ḡ-hé L-H	-	-	-	-	-	-
NC-02X	23								

401

402 The expected outcome for *H-L involving multisyllabic stems is as with HEAD in (25): L-
 403 HL in Edo and H-LL in those NC languages which retain the bisyllabic stem structure. At the same
 404 time, a number of cognates pattern as in NC-02B CHICKEN, where the reflex is H-HL. While it
 405 is reasonable that this should be a distinct correspondence set – perhaps a reconstructed *H-HL
 406 value – we nonetheless collapse it with NC-02A, as we shall see that they pattern alike in our cross-
 407 Edoid comparison (§5). Moreover, a number of cognates have an unexpected all-low pattern,
 408 which we divide in this table based on whether this occurs in Edo (NC-02C), Esan (NC-02D),
 409 Emai (NC-02E), *etc.* (NC-02F through NC-02I, not shown, have five or fewer members – see the
 410 supplemental materials).

411 At the same time, we can also establish a tonal correspondence set where the initial H in Edo
 412 seems not to have shifted, where all NC languages begin with a prefixal H-. This is correspondence
 413 set NC-03 which we reconstruct as a *H-LH melody, in (26). This is split into two equally
 414 numerous subsets: one with an initial H followed by a L in the stem somewhere with no additional

415 H tone (NC-03A), versus one with a H-L plus an additional H in one or more NC languages (NC-
 416 03B). The first subset NC-03A mostly consists of two-syllable nouns (i.e. monosyllabic stems),
 417 while the second NC-03B entirely consists of three-syllable ones.

418 (26) Reflexes of NC-03 *H-LH ($n=54$)

Subset	$n=$	Cognate	bin	ish	ema	ets	une	aaa	oso
NC-03A	18	STONE_1	<i>ú-dò</i> H-L	<i>ú-dò</i> H-L	<i>ú-dò</i> H-L	<i>ú-dò</i> H-L	<i>ú-dò</i> H-L	<i>ū-dò</i> M-L	-
		TONGUE	<i>á-ṛǎṁě</i> H-HL	<i>[úkp]é-làβě</i> [H]H-LL	<i>ó-ěmĩ</i> H-LL	<i>ó-lěmhì</i> H-LL	<i>é-řěmhì</i> H-LL	<i>ȳī-lě</i> M-L	<i>í-rěṛě</i> H-LL
NC-03B	19	COCOYAM	<i>í-jòxó</i> H-LH	<i>í-jòxô</i> H-LF	<i>í-xuòxuó</i> H-LH	<i>í-jàxô</i> H-LF	<i>í-jòxô</i> H-LF	-	-
NC-03C	6	ANT_TERMITE_1	<i>é-dǝ</i> H-L	<i>é-dǝ</i> H-L	<i>é-⁺dó</i> H- ⁺ H	<i>ó-dò</i> H-L	<i>í-dòdô</i> H-LF	-	-
NC-03U	2	BAG_2	<i>é-kpò</i> H-L	-	-	-	-	-	-
NC-03X	9								

419

420 The H-LH reflex of NC-03B suggest that the Great Edo Tone Shift was blocked if the prefixal
 421 H would've become next to a stem H. We may therefore assume that there was a former H tone at
 422 the right edge of words in the *H-LH melody, which became unlinked and ultimately deleted under
 423 certain conditions in NC-03A, such as when there were not enough syllables to host the number
 424 of tones. This final H can account for the non-appearance of the Great Edo Tone Shift in NC-03A,
 425 even though the conditions appear to have been met. This is further supported by a small number
 426 of words (largely from Emai) which show a /H⁺H/ pattern in this set too (NC-03C), which one
 427 could take to be another reflex of the *H-LH melody.

428 The final two NC correspondence sets are the smallest: one reconstructed as *H-H (NC-04)
 429 and one reconstructed as *L-H (NC-05). For the *H-H melody, we take the core reflex to be a H-
 430 H pattern retained only in the Edo language, which in all other NC languages becomes H-L
 431 (lowering of the stem), shown in (27). In some cases, H-H in Edo corresponds to a H-LH pattern

432 (NC-04B) or H-⁺H (NC-04C) in one or more of the other NC languages. Less common are
 433 instances which correspond to an initial L- in one of these other NC languages (NC-04D).

434 (27) Reflexes of NC-04 *H-H (*n*=27)

Subset	<i>n</i> =	Cognate	bin	ish	ema	ets	une	aaa	oso
NC-04A	6	WING	<i>í-fwé</i> H-H	<i>í-hwé</i> H-L	<i>í-fwà</i> H-L	<i>í-fwà</i> H-L	<i>[áb]ì-fò</i> [H]L-L	-	<i>í-fwà</i> H-L
NC-04B	3	NAIL_1	<i>é-hjé</i> H-H	<i>é-hjèlè</i> H-LL	<i>é-hìé</i> H-LH	<i>é-fjè</i> H-L	<i>é-fjèrì</i> H-LL	<i>ē-ēfwè</i> M-ML	<i>é-fjà</i> H-L
NC-04C	7	SHELL	<i>í-γó</i> H-H	<i>í-⁺γô</i> H- ⁺ F	<i>é-⁺γó</i> H- ⁺ H	<i>[íkɸ]í-γó</i> [H]H-H	<i>[íkɸ]í-γò</i> [H]H-L	<i>ē-γō</i> M-M	-
NC-04D	6	COURT_ CASE	<i>é-zó</i> H-H	-	<i>è-zó</i> L-H	<i>è-dzô</i> L-F	<i>è-zô</i> L-F	-	-
NC-04U	2	SWORD_2	<i>á-dá</i> H-H	-	-	-	-	-	-
NC-04X	3								

435
 436 Such patterns suggest that while H-H is faithfully retained in Edo, there was a systematic change
 437 throughout NC (though whose output was conditioned by indeterminate factors at this point). Such
 438 a change accounts for the fact that H-H is systematically banned as a lexical tone pattern in all but
 439 Edo in this Edoid branch: there are no examples of /H-H/ words in Esan, Emai, or Ghotuo (and
 440 only two M-M), only one /H-H/ in Etsako and Uneme, and only two in Ososo.

441 Lastly, the reconstructed *L-H melody is exemplified in (28). In these data, all NC languages
 442 begin with a L-H sequence, with or without an additional low tone word-finally (NC-05A). A
 443 minority of cognates show a downstepped high in Emai (NC-05B), an all-low pattern in one or
 444 more NC languages (NC-05C), or some other unexpected value such as an initial H (NC-05X).

445 (28) Reflexes of NC-05 *L-H ($n=27$)

Subset	$n=$	Cognate	bin	ish	ema	ets	une	aaa	oso
NC-05A	12	HOE_1	è-gwé L-H	è-gwê L-F	è-gúé L-HH	è-gúè L-HL	è-gúè L-HL	è-gbā L-M	è-gúé L-HH
NC-05B	4	BLOOD_2	è-ságjè L-HL	-	è-sád3è L-HL	-	-	ò-òdè L-LL	ò-zè L-L
NC-05C	2	CROCODILE_4	è-yúyù L-HL	-	è-yú ⁺ yú ⁺ L-H ⁺ H	-	-	-	-
NC-05U	6	HOUSE_5	-	-	-	ò-dî L-F	-	-	-
NC-05X	3								

446

447 We summarize the five reconstructed tone classes for NC in (29).

448 (29) Five major correspondence sets in North-Central Edoid (total $n=377$)

P-NC	Set	bin	ish	ema	ets	une	aaa	oso	$n=$	Percentage
*L-L	NC-01	L-L/H-L	L-L	L-L	L-L	L-L	L-L	L-L	121	32.1%
*H-L	NC-02	L-H(L)	H-L	H-L	H-L	H-L	M-L	H-L	119	31.6%
*H-LH	NC-03	H-L(H)	H-L(H)	H-L(H)	H-L(H)	H-L(H)	M-L	H-L(H)	53	14.1%
*H-H	NC-04	H-H	H-L(H)	H-L(H)	H-L(H)	H-L(H)	M-L	H-L(H)	27	7.2%
*L-H	NC-05	L-H(L)	L-H(L)	L-H(L)	L-H(L)	L-H(L)	L-M(L)	L-H(L)	27	7.2%
Other									30	8.0%

449

450 4.4 North-West Edoid (NW)

451 Finally, the North-West branch of Edoid is in the far north of the Edoid cultural complex, an area
 452 with rolling hills closer to the posited homeland for the Proto-Edoid people (Elugbe 1979; Lewis
 453 2013) and even Proto-Benue-Congo itself (Williamson 1989). Of the four Edoid branches, we have
 454 the sparsest data on the North-West branch. We present data from 6 languages: North Ibie [atg],
 455 Oloma [olm], Okpamheri [opa], Uhami [uha], Ehueun [ehu], and Ukue [uku].

456 In general, NW Edoid is remarkable for its lack of tone classes. Most nouns can be placed in
 457 one of two correspondence sets. The first is reconstructed as the now familiar *L-L melody (NW-
 458 01), which in all but one NW language is retained as L-L, in (30). The only location of significant
 459 variation is in Oloma, which for most cognates are transcribed with an additional high at the left

edge resulting in a falling tone on the prefix (NW-01A). The addition of a high tone to an all-low pattern is parallel to that seen already in Degema (§4.1), but where a H was added to the right edge. Even in Oloma, however, there are still tokens transcribed with L-L (NW-01B), or where an Oloma cognate is absent and sorting into one of the two subsets cannot be determined (NW-01U). It is unclear at this point whether this is a synchronic process (as in Degema) or not.

(30) Reflexes of NW-01 *L-L ($n=56$)

Subset	$n=$	Cognate	atg	olm	opa	uha	ehu	uku
NW-01A	20	TOOTH	à-kò L-L	lê-kò F-L	rà-kò L-L	à-kò L-L	à-kũ L-L	à-kũ L-L
NW-01B	6	CHIN	è-kpàmi L-LL	à-gbà L-L	à-gbà L-L	à-gbàmi L-LL	à-wǎ L-L	à-gbǎ L-L
NW-01U	20	SOIL_1	è-kè L-L	-	è-kè L-L	è-kè L-L	è-kè L-L	è-kè L-L
NW-01X	10							

The second correspondence NW-02 is reconstructed as a *H-L melody, in (31). The reflex of this class is H-L in all NW languages except Ehueun where it is mostly M-L (NW-02A) but occasionally H-L (NW-02B). There exist a small number of subsets involving an all-low pattern in one or more NW language (NW-03C), a rising word-level contour (e.g. L-H or M-H – NW-03D), or a H-LH pattern (NW-03E).

472 (31) Reflexes of NW-02 *H-L ($n=103$)

Subset	$n=$	Cognate	atg	olm	opa	uha	ehu	uku
NW-02A	25	HEAD	<i>ú-xòmhì</i> H-LL	<i>ghí-kà</i> H-L	<i>ú-khèmhì</i> H-LL	<i>ú-kèmi</i> H-LL	<i>ū-hǔ</i> M-L	<i>ú-kòmi</i> H-LL
NW-02B	4	MOUTH	-	<i>ú-nù</i> H-L	<i>ú-nù</i> H-L	<i>ú-nù</i> H-L	<i>ú-nù</i> H-L	<i>ú-nù</i> H-L
NW-02C	15	CORPSE_2	-	-	<i>ó-khùù</i> H-LL	<i>ò-gù</i> L-L	<i>ō-gǔ</i> M-L	<i>ó-gǔ</i> H-L
NW-02D	8	NIGHT_1	-	<i>ghá-úzé</i> H-HL	<i>á-zò</i> H-L	<i>è-dùzí</i> L-LH	<i>ū-dǔí</i> M-H	<i>á-rǔ</i> H-L
NW-02E	3	SNAIL	<i>ú-lùé</i> H-LH	<i>ú-thùnhámi</i> H-LHL	-	-	-	-
NW-02U	41	MOTHER_1	-	-	-	-	<i>ō-nǔ</i> M-L	-
NW-02X	7							

473

474 We summarize the two reconstructed tone classes for NW in (32). The “other” row are those
 475 which do not fit into these correspondence sets, the largest of these being a tentative set of L-H
 476 nouns which appear in one or more NW language (e.g. HOUSE_5 *òdé* in Uhami and Ukue, and
 477 *ōdé* in Ehueun).

478 (32) Two major correspondence sets in North-West Edoid (total $n=186$)

P-NC	Set	atg	olm	opa	uha	ehu	uku	$n=$	Percentage
*L-L	NW-01	L-L	F-L/L-L	L-L	L-L	L-L	L-L	56	30.1%
*H-L	NW-02	H-L	H-L	H-L	H-L	M-L	H-L	103	55.4%
Other								27	14.5%

479

480 4.5 Summary of proposed tone changes

481 We summarize our correspondence sets, proto-melodies, and descendent reflexes in (33). Tone
 482 changes from the proto-stage to the individual branches are in red.

483 (33) All major correspondence sets across Edoid and their common reflexes (changes in red)

P-DE set			deg	enn	epi				
a.	DE-01	*L-L	> L- H	L-L	L-L				
b.	DE-02	*H-H	> H- [↓] H	H- L	L-HM/H-HM				
c.	DE-03	*H-L	> H- HL	H-L	L-HM				
d.	DE-04	*L-H	> L-H [↓] H	L- HL	L-HM				
P-SW set			erh	iso	oke	urh	evh		
e.	SW-01	*L-L	> L- H /L-L	L-L	H-L /L-L	L-L	L-L		
f.	SW-02	*L-LH	> L- H	L- L	L-H	L-L	L-L		
g.	SW-03	*L-H	> L-H	L-H	L-H	L-H	L-H		
h.	SW-04	*H-L	> H-L	H-L	H-L	H-L/H- M	H-L		
i.	SW-05	*H-H	> H- L	H-H	L-H	H-H	L-H		
P-NC set			bin	ish	ema	ets	une	aaa	oso
j.	NC-01	*L-L	> L-L/ H-L	L-L	L-L	L-L	L-L	L-L	L-L
k.	NC-02	*H-L	> L-H (L)	H-L	H-L	H-L	H-L	M-L	H-L
l.	NC-03	*H-LH	> H-L(H)	H-L(H)	H-L(H)	H-L(H)	H-L(H)	M-L	H-L(H)
m.	NC-04	*H-H	> H-H	H- L (H)	H- L (H)	H- L (H)	H- L (H)	M-L	H- L (H)
n.	NC-05	*L-H	> L-H(L)	L-H(L)	L-H(L)	L-H(L)	L-H(L)	L- M (L)	L-H(L)
P-NW set			atg	olm	opa	uha	ehu	uku	
o.	NW-01	*L-L	> L-L	F-L /L-L	L-L	L-L	L-L	L-L	
p.	NW-02	*H-L	> H-L	H-L	H-L	H-L	M-L	H-L	

484

485 We can group tone changes into various types. First, independently in at least three Edoid
 486 branches there has been a rule which adds a high tone to an all-low noun, what we can call “H-
 487 Obligatoriness” (using the term from Hyman 2006). In Delta Edoid, *L-L becomes L-H in Degema
 488 in words in isolation, a synchronically active rule in the language (Rolle & Kari 2022). In South-
 489 West Edoid, *L-L becomes L-H in Eruwa and H-L in Okpe in a great many cognates (though
 490 others remain L-L), and in Oloma in North-West Edoid most *L-L words have become F-L. The
 491 status of these latter changes as synchronic or purely diachronic is not presently known.

492 Another type of recurring change is lowering a H tone to a downstepped [↓]H, a mid M, or a
 493 low L, referred to as “H-Lowering”. In some instances, all H’s lower to M regardless of position,
 494 such as in the NC language Ghotuo [aaa] or the NW language Ehueun, likely due to the
 495 development of a higher contrastive pitch. In other cases, H-lowering took place only after a H

tone, whether to downstep (Degema), mid (Epie-Atissa), or low (Engenni; Eruwa; almost all of NC Edoid). The reverse process of “L-Raising” is much less common, and is found only in Urhobo where *H-L becomes either H-L or H-M. Even here, it can be debated whether this is an innovation or a retention, and whether this is a synchronic or purely diachronic alternation.

Another common change is “Rightward H-Shift”, best exemplified by what we called the Great Edo Tone Shift whereby Proto-NC *H-L becomes L-H(L) in Edo [bɪn]. Comparable shifts are also implicated by the change from Proto-DE *H-H and *H-L to Epie-Atissa L-HM, and from Proto-SW *H-H to Eruwa and Uvwie [evh] L-H. One could even understand the Degema change from *H-L to H-HL as of this type, albeit as spreading rather than shifting.

The last major type of change is found in South-West Edoid, where Proto-SW *L-LH changed in all languages: to L-H in Eruwa and Okpe but to L-L in the others. Rising contours within stems are marked in Edoid (clearly dispreferred and in fact nearly absent in DE and NW), and are entirely eliminated after L in SW by expanding either the low or high portion. We attribute this change to a type of constraint we call “Rising-Markedness”, where a rising sequence in a stem is dispreferred after a low prefix.

5 Reconstructing Proto-Edoid tone

We can now present our reconstruction of Proto-Edoid (PE) tone using these intermediate reconstructions. By systematically comparing our intermediate reconstructions across the four branches, we are able to determine eight correspondence sets, neutrally labeled classes PE-01 to PE-08 in (34). These constitute 312 of the total 436 cognates (71.6%). In addition, there are a number of cognates which are ambiguous as to which set they belong to (PE+, labeled “ambiguous” – $n=88$), as well as many which are simply irregular (PE-00, labeled “other” – $n=36$).

The PE classes and their reflexes in (34) are grouped according to where there have been noun class melody mergers (mergers are boxed). At the geographic extremes of Edoid – recall the map in (3) – P-DE and P-NW each have drastically reduced the Proto-Edoid system, with the resultant tone pattern primarily determined by the tone value of the prefix in Proto-Edoid. In contrast, P-SW and P-NC in the interior of the Edoid zone are more conservative while still also showing class mergers largely based around whether there is a single H in the word or two H's (one on the prefix and one on the stem).

(34) Summary of PE proposal with reflexes in intermediate branches (mergers are boxed)

Branch				Reconstruction			Cognate confidence			Of total		
P-DE	P-SW	P-NC	P-NW	and PE-set		<i>n</i> =	(	High	Med.	Low	)	(<i>n</i> =436)
*L-L	*...-L	*L-L	*L-L	**L-L	PE-01	113	(	62	37	14	)	25.9%
*L-L	*L-H	*L-H	*L-L	**L-LH	PE-02	27	(	9	14	4	)	6.2%
*L-L	*L-H	*H-L	*H-L	**H-L	PE-05	20	(	7	13	0	)	4.6%
*L-L	*L-LH	*H-L	*H-L	**L-H	PE-03	38	(	21	15	2	)	8.7%
?	*L-L	*H-L	*H-L	**L-HL	PE-04	13	(	7	6	0	)	3.0%
*H-...	*H-L	*H-L	*H-L	**H-HL	PE-07	15	(	5	10	0	)	3.4%
*H-...	*H-L	*H-LH	*H-L	**H-LH	PE-06	54	(	26	21	7	)	12.4%
*H-...	*H-...	*H-...	*H-L	**H-H	PE-08	32	(	16	11	5	)	7.3%
				Ambiguous	PE+	88	(	25	26	37	)	20.2%
				Other	PE-00	36	(	12	6	18	)	8.3%

For their eight PE-sets, we reconstruct a two-way H/L contrast on the noun class prefix (PE-01 to PE-04 vs. PE-05 to PE-08), and a distinction between H/L/HL/LH on stems (recall that here and throughout, two ** mean PE). The number of members in each set is quite uneven, ranging from over 25% for PE-01 **L-L (n=113) to only 3% for PE-04 **L-HL (n=13). There is a fairly even distribution between cognates with high confidence (matching a pre-established consonant correspondence) and those with medium confidence (having a plausible but non-established consonant correspondence); see §3. Notably, in the four most common proto-melodies – **L-L,

534 **L-H, **H-LH, and **H-H – there are more high than medium members. Cognates of low
 535 confidence are those with only reflexes in a single branch, and are largely in the ambiguous
 536 category.

537 In this section, we first discuss **L-L and **L-LH with uniform L-initial reflexes (what we
 538 call the “low” grouping – §5.1), then **H-L, **L-H, and **L-HL with both H- and L-initial
 539 reflexes (the “high/low” grouping – §5.2), then **H-HL, **H-LH, and **H-H with only H-initial
 540 reflexes (the “high” grouping – §5.3). To close this section, we provide justification for these
 541 specific reconstructions and how they are linked together (§5.4).

542 5.1 The low grouping

543 The low grouping consists of two correspondence sets, one reconstructed as PE **L-L (PE-01)
 544 and one as **L-LH (PE-02). The former set is by far the largest in PE ($n=113$), twice the size of
 545 the next largest set. Just as we reconstructed major and minor subsets for each correspondence set
 546 for the four individual Edoid branches, we also do so for PE, shown in (35). The majority of **L-
 547 L reflexes show *L-L across the four branches (as in PE-01A), but a notable minority show *H-L
 548 in P-SW (PE-01B). Others are ambiguous because a P-SW form is incidentally missing (PE-01U),
 549 and still others show other irregularities involving an unexpected high somewhere (PE-01X).

550 (35) Reflexes of PE-01 **L-L in four daughter branches ($n=113$)

Subset	$n=$	Cognate	P-DE	P-SW	P-NC	P-NW
PE-01A	46	TOOTH	*L-L	*L-L	*L-L	*L-L
PE-01B	18	ASH	*L-L	*H-L	*L-L	*L-L
PE-01U	31	DOOR_2	*L-L	-	*L-L	*L-L
PE-01X	18					

551

552 Table (36) involves two PE-01A cognates TOOTH and FOOT_LEG_2, showing the expected
 553 reflexes in each language for *L-L (recall H-Obligatoriness changes, as discussed in §4.5 most
 554 recently).

555 (36) PE-01A **L-L cognates TOOTH and FOOT_LEG_2 (with *L-L reflexes)

Branch	Language	TOOTH	Tones	Proto-T	FOOT_LEG_2	Tones	Proto-T
a. DE	deg	ò-kó	L-H	*L-L	à-βí	L-H	*L-L
	enn	à-kò	L-L		à-vì	L-L	
	epi	à-kò	L-L		à-vì	L-L	
b. SW	erh	à-kó	L-H	*L-L	à-uò	L-L	*L-L
	iso	à-kò	L-L	(SW-01A)	ò-wò	L-L	(SW-01C)
	oke	à-wò	H-L		ò-wò	L-L	
	urh	à-kò	L-L		ò-wò	L-L	
	evh	à-kò	L-L		à-uò	L-L	
c. NC	bin	à-kò	L-L	*L-L	ò-wè	L-L	*L-L
	ish	à-kò	L-L		ò-wè	L-L	
	ema	à-kò	L-L		ò-è	L-L	
	ets	è-kò	L-L		ò-wè	L-L	
	une	à-kò	L-L		-	-	
	aaa	èè-kò	L-L		-	-	
	oso	[í]rè-kò	[H]L-L		ò-wè	L-L	
d. NW	atg	à-kò	L-L	*L-L	ò-wè	L-L	*L-L
	olm	lê-kò	F-L		γô-mhì[nà]	F-L[L]	
	opa	rà-kò	L-L		-	-	
	uha	à-kò	L-L		-	-	
	ehu	à-kù	L-L		-	-	
	uku	à-kù	L-L		-	-	

556

557 In contrast, (37) shows PE-01B cognates ASH and FAECES_1 which have a *H-L in P-SW (in
558 red), and consequently have an initial H- in all daughter languages. At this point, there is no
559 obvious conditioning factor determining when PE **L-L becomes P-SW *L-L vs. *H-L; we
560 interpret this as a variable application of H-Obligatoriness in the development of PE to P-SW.

561 (37) PE-01B **L-L cognates ASH and FAECES_1 (exceptional *H-L reflex in P-SW)

Branch	Language	ASH	Tones	Proto-T	FAECES_1	Tones	Proto-T
a. DE	deg	à-mà[fú]	L-L[H]	*L-L	ì-sò	L-H	*L-L
	enn	è-bù[fù]	L-L[L]		-	-	
	epi	ù-wò	L-L		-	-	
b. SW	erh	í-mò	H-L	*H-L	-	-	*H-L
	iso	é-ηwò	H-L	(SW-04A)	í-sò	H-L	(SW-04B)
	oke	-	-		í-sò̃	H-L	
	urh	í-wù[rìè]	H-L[L]		í-sò̃	H-M	
	evh	-	-		í-sò̃	H-L	
c. NC	bin	è-ũè̃	L-L	*L-L	ì-sà̃	L-L	*L-L
	ish	é-vò̃	H-L		-	-	
	ema	è-mùè	L-LL		ì-sò̃	L-L	
	ets	è-ηwèè	L-LL		ì-tsò	L-L	
	une	è-mhò	L-L		ì-sò	L-L	
	aaa	γè-mhò	L-L		ì-sò	L-L	
	oso	è-mwê	L-F		ì-sò	L-L	
d. NW	atg	è-nwè	L-L	*L-L	-	-	*L-L
	olm	ghê-mò	F-L		ì-sò	L-L	
	opa	wè-mò	L-L		ì-sò	L-L	
	uha	è-mò	L-L		í-sò	H-L	
	ehu	è-mò	L-L		ì-sò̃	L-L	
	uku	è-mò	L-L		ì-ĩò̃	L-L	

562

563 The next correspondence set is PE-02 **L-LH ($n=27$), in (38). Its reflex is *L-H in the more
564 centrally located P-SW and P-NC branches but *L-L in the more peripheral P-DE and P-NW
565 branches. The expected reflexes are subset PE-02A, such as WALKING_STICK expanded in (39)
566 with modern reflexes. Additional subsets include those with an unexpected *L-L reflex in P-SW
567 (PE-02B, e.g. BLOOD_2) or *L-L in P-NC (PE-02C). As apparent from the example cognates in
568 (39), this correspondence set is marked by fewer reflex forms in individual languages. As a whole,
569 **L-LH is dwarfed by the number of **L-L cognates (more than four times the size), and the
570 majority of *L-LH cognates are of medium confidence while the majority of *L-L are of high
571 confidence (see (34)).

572 (38) Reflexes of PE-02 **L-LH ($n=27$)

Subset	$n=$	Cognate	P-DE	P-SW	P-NC	P-NW
PE-02A	11	WALKING_STICK	*L-L	*L-H	*L-H	*L-L
PE-02B	4	BLOOD_2	*L-L	*L- L	*L-H	*L-L
PE-02C	3	PROVERB	*L-L	*L-H	*L- L	*L-L
PE-02U	4	COWIFE_1	-	-	*L-H	-
PE-02X	5					

573

574 (39) Reflexes of PE **L-LH – WALKING_STICK (PE-02A) and BLOOD_2 (PE-02B)

Branch	Language	WALKING STICK	Tones	Proto-T	BLOOD 2	Tones	Proto-T
a. DE	deg	<i>ò-kpòtí</i>	L-LH	*L-L	<i>ì-zàlà</i>	L-LL	*L-L
	enn	-	-		-	-	
	epi	-	-		<i>ì-zàrà</i>	L-LL	
b. SW	erh	-	-	*L-H	<i>à-zí</i>	L-H	*L-L
	iso	-	-		<i>à-zè</i>	L-L	
	oke	-	-		-	-	
	urh	<i>ò-kpó</i>	L-H		-	-	
	evh	-	-		-	-	
c. NC	bin	<i>ò-kpá</i>	L-H	*L-H	<i>è-ságjě</i>	L-HL	*L-H
	ish	<i>[úkp]ò-kpó</i>	[H]L-H		-	-	
	ema	<i>ó-⁺kpó</i>	H- ⁺ H		<i>è-sádžě</i>	L-HL	
	ets	-	-		-	-	
	une	-	-		-	-	
	aaa	-	-		<i>ò-òdè</i>	L-LL	
	oso	-	-		<i>ò-zè</i>	L-L	
d. NW	atg	-	-		-	-	*L-L
	olm	-	-		<i>â-zè</i>	F-L	
	opa	-	-		<i>à-dzè</i>	L-L	
	uha	-	-		<i>è-zè</i>	L-L	
	ehu	-	-		<i>è-rè</i>	L-L	
	uku	-	-		<i>è-dè</i>	L-L	

575

576 The change from PE **L-LH to *L-L or *L-H (with variability in some cognates) directly
577 mirrors the behavior of *L-LH in SW we saw in §4.2, whose reflex was L-L or L-H depending on
578 the language. We turn next to the origin of SW's *L-LH melody, where we analyze it as from PE
579 **H-L.

580 **5.2 The high/low grouping**

581 Next is the high/low grouping, consisting of three correspondence sets: **L-H (PE-03), **L-HL
582 (PE-04), and **H-L (PE-05). These three have in common that they show both a H- and a L- prefix
583 in the four descendent proto-languages, whereas the other groupings show a consistent initial L-
584 or initial H- reflex. These proto-melodies also share the fact that they have an opposite value at the
585 internal stem boundary, which will become relevant later (§5.4).

586 Starting with PE-03 **L-H (*n*=38), this becomes *L-L in P-DE, *L-LH in P-SW, and *H-L
587 in P-NC and P-NW, shown in (40). Two cognates are provided in (41), one with the expected
588 reflex (HEAD – PE03A) and one with irregularity (in this example, concentrated within the DE
589 branch for HOUSE_1 – PE03X).

590 (40) Reflexes of PE-03 **L-H (*n*=38)

Subset	<i>n</i> =	Cognate	P-DE	P-SW	P-NC	P-NW
PE-03A	24	HEAD	*L-L	*L-LH	*H-L	*H-L
PE-03B	2	URINE_1	*L-L	*L- L	*H-L	*H-L
PE-03U	2					
PE-03X	10					

592 (41) Reflexes of PE **L-H – HEAD (PE-03A) and HOUSE_1 (PE-03X)

Branch	Language	HEAD	Tones	Proto-T	HOUSE_1	Tones	Proto-T
a. DE	deg	ù-tóm	L-H	*L-L	ó- ⁺ βáj	H- ⁺ H	*H-H
	enn	ù-tòmù	L-LL		-	-	
	epi	ù-tòm	L-L		ò-wòò	L-LL	
b. SW	erh	ù-zòvú	L-LH	*L-LH	ò-wá	L-H	*L-LH
	iso	ù-zòù	L-LL		-	-	
	oke	ù-rómú	L-HH		ò-γwá	L-H	
	urh	ù-jòvù	L-LL		ù-wèvi	L-LL	
	evh	-	-		ù-γwòmù	L-LL	
c. NC	bin	ù-húmù	L-HL	*H-L	ò-wá	L-H	*H-L
	ish	ù-hòβò	H-LL		ù-wâ	H-F	
	ema	ù-hùmì	H-LL		ó-à	H-L	
	ets	ù-sò	H-L		ó-và	H-L	
	une	ù-sòmhi	H-LL		ó-và	H-L	
	aaa	ù-sò	M-L		ò-vàγi	M-LL	
	oso	ù-kwè	H-L		ó-wà	H-L	
d. NW	atg	ù-xòmhi	H-LL	*H-L	ó-wà	H-L	*H-L
	olm	γí-kà	H-L		-	-	
	opa	ù-xèmhi	H-LL		-	-	
	uha	ù-kèmì	H-LL		-	-	
	ehu	ù-hù	M-L		-	-	
	uku	ù-kòmì	H-LL		-	-	

593

594 For these melodies, PE is reconstructed with an initial L- (in **L-H) and the P-NC and P-NW
 595 branches show an initial H-. We must therefore posit an “externalization” process which moved
 596 stem tone to the prefix position. At the same time, we also posit that the stem high of **L-H was
 597 either lost (in DE *L-L) or moved rightward (in SW *L-LH).

598 Next consider PE-04 reconstructed as **L-HL, in (42). This has a *H-L reflex in both P-NC
 599 and P-NW, showing the same externalization process as with PE-03 **L-H in (41). However,
 600 unlike there, P-SW is *L-L (rather than *L-LH). Interestingly, P-DE shows variation between *L-
 601 L (PE-04A) and *H-H reflexes (PE-04B), though one should note the very low number of members
 602 in this correspondence set ($n=13$).

603 (42) Reflexes of PE-04 **L-HL ($n=13$)

Subset	$n=$	Cognate	P-DE	P-SW	P-NC	P-NW
PE-04A	4	GREY_HAIR	*L-L	*L-L	*H-L	*H-L
PE-04B	3	DOG_1	*H-H	*L-L	*H-L	*H-L
PE-04U	3					
PE-04X	3					

604

605 (43) Reflexes of PE **L-HL – GREY_HAIR (PE-04A) and DOG_1 (PE-04B)

Branch	Language	GREY_HAIR	Tones	Proto-T	DOG_1	Tones	Proto-T
a. DE	deg	ò-dáŋ	L-H	*L-L	á- ⁺ bówá	H- ⁺ HH	*H-H
	enn	è-dàà	L-LL		á-bùà	H-LL	
	epi	ì-dàn	L-L		à-búā	L-HM	
b. SW	erh	-	-	*L-L	-	-	*L-L
	iso	-	-		è-bò	L-L	
	oke	é-ṙà	H-L		-	-	
	urh	è-fà	L-L		-	-	
	evh	-	-		-	-	
c. NC	bin	è-dé	L-H	*H-L	à-wá	L-H	*H-L
	ish	-	-		á-wà	H-L	
	ema	é-dè	H-L		á-wà	H-L	
	ets	é-dè	H-L		á-ywà	H-L	
	une	é-dè	H-L		á-wà	H-L	
	aaa	ghō-dè	M-L		yā-wà	M-L	
	oso	-	-		á-wà	H-L	
d. NW	atg	é-dè	H-L	*H-L	à-wó[ŋ]	L-H[H]	*H-L
	olm	ghó-zè	H-L		ghá-wà	H-L	
	opa	-	-		wá-bùà	H-LL	
	uha	-	-		à-bùà	L-LL	
	ehu	-	-		ē-bò	M-L	
	uku	-	-		é-bò	H-L	

606

607 The final high/low correspondence set is PE-05 reconstructed as **H-L, in (44). In this set,

608 P-NC and P-NW remain *H-L, while P-SW becomes *L-H and P-DE becomes *L-L (for the most

609 part).

610 (44) Reflexes of PE-05 **H-L ($n=20$)

Subset	$n=$	Cognate	P-DE	P-SW	P-NC	P-NW
PE-05A	12	TORTOISE_1	*L-L	*L-H	*H-L	*H-L
PE-05X	8					

611

612 (45) Reflexes of PE **H-L – TORTOISE_1 (PE-05A) and ANTELOPE_1 (PE-05X)

	Branch	Language	TORTOISE_1	Tones	Proto-T	ANTELOPE_1	Tones	Proto-T
a.	DE	deg	<i>è-wíj</i>	L-H	*L-L	<i>é-túwèj</i>	H-HL	*H-L
		enn	<i>è-wì</i>	L-LL		<i>é-tùè</i>	H-LLL	
		epi	<i>è-wì</i>	L-L		<i>ò-túè</i>	L-HFL	
b.	SW	erh	-	-	*L-H	<i>è-rúé</i>	L-HH	*L-H
		iso	<i>ò-gbé</i>	L-H		<i>è-rúé</i>	L-HH	
		oke	-	-		<i>è-rúé</i>	L-HH	
		urh	<i>ò-gbéjĩ</i>	L-HM		<i>è-rúé</i>	L-HH	
		evh	-	-		<i>è-rúé</i>	L-HH	
c.	NC	bin	<i>è-gúí</i>	L-HH	*H-L	<i>é-rùè</i>	L-LL	*H-L
		ish	<i>é-wì</i>	H-L		<i>é-rùè</i>	H-LL	
		ema	<i>é-ĩ</i>	H-L		-	-	
		ets	<i>é-yì</i>	H-L		<i>é-rwè</i>	H-L	
		une	<i>é-hù</i>	H-L		<i>é-θwè</i>	H-L	
		aaa	<i>ē-ũ</i>	M-L		<i>ē-rùè</i>	M-LL	
		oso	<i>é-gù</i>	H-L		-	-	
d.	NW	atg	-	-	*H-L	-	-	*H-L
		olm	-	-		<i>á-tù</i>	H-L	
		opa	<i>é-gùì</i>	H-LL		-	-	
		uha	<i>é-gì</i>	H-L		-	-	
		ehu	<i>ē-gùè</i>	M-LL		<i>è-rùè</i>	L-LL	
		uku	<i>é-gùì</i>	H-LL		<i>é-tùè</i>	H-LL	

613

614 The crucial aspect of the PE-05 set is that while the **H-L value remains as such in P-NC and
615 P-NW, there is a rightward shift of the H tone in P-SW. This is analogous to other rightward shifts
616 already seen, e.g. the Great Edo Tone Shift (§4.3).

617 5.3 The high grouping

618 Lastly, the high grouping consists of three correspondence sets whose reflexes show an initial H-
619 in all four intermediate proto-languages. These are **H-LH (PE-06), **H-HL (PE-07), and **H-
620 H (PE-08), which have in common that they have a H both on the prefix and somewhere within

the stem. Before we detail these sets, we must admit that this grouping is perhaps the most tentative due to frequent subsets with unclear conditioning. This is particularly true of the Delta branch, where these three proto-melodies are not consistent as to whether they become P-DE *H-L or *H-H. We collapse these as *H-L/*H-H in the tables to follow (and denoted as “*H-...” in (34) above).

Beginning with PE-06 reconstructed as **H-LH, this remains *H-LH in P-NC and becomes *H-L in P-NW, in (46). In P-SW, however, we see most significant variation: while most P-SW reflexes of this set are *H-L (PE-06A), a minority are *L-H (PE-06B) or *L-L (PE-06C). Still others are either ambiguous in its subset (PE-06U, $n=12$) or cannot be fit into these subsets (PE-06X, $n=19$), numbers comparable to the main subset PE-06A ($n=14$). For P-DE, as stated it becomes either *H-L or *H-H, with too much variation to privilege one over the other.

(46) Reflexes of PE-06 **H-LH ($n=54$)

Subset	$n=$	Cognate	P-DE	P-SW	P-NC	P-NW
PE-06A	14	SNAIL	*H-L/*H-H	*H-L	*H-LH	*H-L
PE-06B	5	GRASSCUTTER_3	*H-L/*H-H	*L-H	*H-LH	*H-L
PE-06C	4	BOUNDARY_1	*H-L/*H-H	*L-L	*H-LH	*H-L
PE-06U	12					
PE-06X	19					

633 (47) Reflexes of PE **H-LH – SNAIL (PE-06A) and BOUNDARY_1 (PE-06C)

	Branch	Language	SNAIL	Tones	Proto-T	BOUNDARY_1	Tones	Proto-T
a. DE	deg	-	-	-		ó- ⁺ <i>gúrú</i>	H- ⁺ HHH	*H-H
	enn	-	-	-		-	-	
	epi	-	-	-		-	-	
b. SW	erh	-	-	*H-L	-	-	-	*L-L
	iso	ù-sè[<i>kpè</i>]	L-L[L]		-	-	-	
	oke	-	-		-	-	-	
	urh	ú-sè[<i>kpè</i>]	H-L[L]		ù- <i>ywrù</i>	L-L		
	evh	-	-		-	-	-	
c. NC	bin	ú- <i>rèé</i>	H-LH	*H-LH	ú- <i>yù</i>	H-L		*H-LH
	ish	ú- ⁺ <i>rê</i>	H- ⁺ F		ú- <i>yù</i>	H-L		
	ema	ú- ⁺ <i>ré</i>	H- ⁺ H		-	-		
	ets	ú- <i>rě</i>	H-R		-	-		
	une	ú- <i>thèi</i>	H-LL		-	-		
	aaa	ū- <i>ré</i>	M-H		-	-		
	oso	-	-		-	-		
d. NW	atg	ú-lùé	H-LH	*H-L	-	-		
	olm	ú-thù[<i>nhámhi</i>]	H-L[HL]		-	-		
	opa	-	-		-	-		
	uha	-	-		-	-		
	ehu	-	-		-	-		
	uku	-	-		-	-		

634

635 In contrast, PE-07 is much smaller ($n=15$), reconstructed as **H-HL. P-DE and P-NW
636 branches have identical reflexes as with PE-06 **H-LH above, namely *H-L/*H-H and *H-L
637 respectively. The key differences are in P-SW which consistently shows a *H-L pattern, and P-
638 NC which shows *H-L rather than *H-LH. This is evidenced most clearly by the fact that reflexes
639 of P-NC *H-L (< PE-07 **H-HL) in Edo undergo the Great Edo Tone Shift, while reflexes of P-
640 NC *H-LH (< PE-06 **H-LH) do not.

641 (48) Reflexes of PE-07 **H-HL ($n=15$)

Subset	$n=$	Cognate	P-DE	P-SW	P-NC	P-NW
PE-07A	9	RIVER	*H-L/*H-H	*H-L	*H-L	*H-L
PE-07X	6					

642

643 (49) Reflexes of PE **H-HL – RIVER (PE-07A) and ROPE_1 (PE-07X)

	Branch	Language	RIVER	Tones	Proto-T	ROPE_1	Tones	Proto-T
a.	DE	deg	é- ⁺ dá	H- ⁺ H	*H-H	ù-díŋ ^w	L-H	*L-L
		enn	é-dà	H-L		ì-dìì	L-LL	
		epi	é-dē	H-M		ù-dìì	L-LL	
b.	SW	erh	í-sè	H-L	*H-L	-	-	*H-L
		iso	é-t̥è	H-L		-	-	
		oke	-	-		-	-	
		urh	-	-		ú-rùr̥é	H-LH	
		evh	-	-		-	-	
c.	NC	bin	è-zè	L-L	*H-L	ì-rí	L-H	*H-L
		ish	é-dè	H-L		[úkp]í-rì	[H]H-L	
		ema	é-dà	H-L		ú-ì	H-L	
		ets	é-dà	H-L		ú-lì	H-L	
		une	é-dò	H-L		ú-lì	H-L	
		aaa	ē-dà	M-L		ū-ùγì	M-LL	
		oso	-	-		ú-rídià	H-HL	
d.	NW	atg	-	-	*H-L	-	-	*H-L
		olm	-	-		ghú-rìghì	H-LL	
		opa	-	-		ú-rì	H-L	
		uha	-	-		ú-dì	H-L	
		ehu	-	-		ù-rī	L-M	
		uku	-	-		-	-	

644

645 As with PE-06 **H-LH, the prefix H- in the PE-07 **H-HL melody does not shift rightward
 646 in P-SW (hence the reconstructed value *H-L for both RIVER and ROPE_1 in (49)). This contrasts
 647 with PE-05 **H-L, where it does shift into the stem in P-SW (resulting in *L-H, e.g. (45)). The
 648 key difference is that the PE melodies **H-LH and **H-HL have a stem H which prevents H-tone
 649 shift whereas the latter **H-L melody does not.

650 The last set is PE-08 reconstructed as **H-H. This has the most subsets and exhibits frequent
 651 variation and irregularity, which may suggest that it actually constitutes more than one proto-
 652 melody which is no longer distinguishable (at least not at this time). One impetus for reconstructing
 653 a **H-H set in PE is to account for the presence of H-H reflexes in *SW and *NC which have not
 654 yet been shown to be reflexes of any PE category. The largest subset is PE-08A, which has a *H-

655 H reflex in P-SW, *H-L in P-NC and P-NW, and variably *H-H/*H-L in P-DE. At the same time,
 656 there are also a few examples of P-NC *H-H which corresponds either to P-SW *H-H (PE-08B)
 657 or *H-L (PE-08C). Note however the large number of irregular patterns in PE-08X ($n=12$), twice
 658 as much as PE-08A. This melody PE-08 **H-H stands as the most tentative PE reconstruction.
 659 (50) Reflexes of PE-08 **H-H ($n=32$)

Subset	$n=$	Cognate	P-DE	P-SW	P-NC	P-NW
PE-08A	6	EAR	*H-L/*H-H	*H-H	*H-L	*H-L
PE-08B	2	NOSE	*H-L/*H-H	*H-H	*H-H	*H-L
PE-08C	3	STALL	*H-L/*H-H	*H-L	*H-H	*H-L
PE-08U	9					
PE-08X	12					

660

661 (51) Reflexes of PE **H-H – EAR and TREE (PE-08A)

	Branch	Language	EAR	Tones	Proto-T	TREE	Tones	Proto-T
a.	DE	deg	ú- [↓] só	H- [↓] H	*H-H	ú- [↓] táŋ	H- [↓] H	*H-H
		enn	é-sòù	H-LL		é-tàì	H-LL	
		epi	í-sóḡ	H-HM		ì-táān	L-HM	
b.	SW	erh	-	-	*H-H	ú-rârê	H-LL	*H-H
		iso	ú-zó	H-H		ú-ré	H-H	
		oke	ò-ró	L-H		ò-rá	L-H	
		urh	ó-ró	H-H		ú-ré	H-H	
		evh	è-só	L-H		ò-rá	L-H	
c.	NC	bin	è-hó	L-H	*H-L	è-rá	L-H	*H-L
		ish	é-hò	H-L		é-rà	H-L	
		ema	é-hḡ	H-L		ó-rà	H-L	
		ets	é-uò	H-L		ó-rà	H-L	
		une	é-sò	H-L		ó-θà	H-L	
		aaa	γō-hòγì	M-LL		ō-tà	M-L	
		oso	ó-sò	H-L		ó-thè	H-L	
d.	NW	atg	é-sò	H-L	*H-L	ó-rè	H-L	*H-L
		olm	ghó-žò	H-L		ó-shà	H-L	
		opa	wó-žò	H-L		ú-fà	H-L	
		uha	é-sò	H-L		ó-fà	H-L	
		ehu	é-rò	H-L		è-fā	L-M	
		uku	é-rò	H-L		ò-fā	L-F	

662

663 (52) Reflexes of PE **H-H – NOSE (PE-08B) and STALL (PE-08C)

	Branch	Language	NOSE	Tones	Proto-T	STALL	Tones	Proto-T
a.	DE	deg	<i>í-súβèŋ</i>	H-HL	*H-L	-	-	
		enn	<i>ú-swèì</i>	H-LL		-	-	
		epi	<i>ù-súḃē</i>	L-HM		-	-	
b.	SW	erh	<i>û-wě̃</i>	F-L	*H-H	-	-	*H-L
		iso	<i>ú-ŋwé</i>	H-H		-	-	
		oke	<i>ì-rúé̃</i>	L-HH		-	-	
		urh	<i>ú-wé̃</i>	H-H		<i>ó-ywà</i>	H-L	
		evh	-	-		-	-	
c.	NC	bin	<i>í-húé</i>	H-HH	*H-H	<i>ó-wá</i>	H-H	*H-H
		ish	<i>í-⁺hwé</i>	H- ⁺ F		<i>ó-wáà</i>	H-HL	
		ema	<i>í-hùè</i>	H-LL		-	-	
		ets	<i>í-sùè</i>	H-LL		<i>ò-gwá[kì]</i>	L-H[L]	
		une	<i>í-fùè</i>	H-LL		<i>é-gbhá[kò]</i>	H-H[L]	
		aaa	<i>ī-rùè</i>	M-LL		-	-	
		oso	<i>í-fwè</i>	H-L		-	-	
d.	NW	atg	-	-	*H-L	-	-	
		olm	<i>ú-žùò</i>	H-LL		-	-	
		opa	<i>ú-sùò</i>	H-LL		-	-	
		uha	<i>í-sù</i>	H-L		-	-	
		ehu	<i>í-rò</i>	H-L		-	-	
		uku	<i>í-rò</i>	H-L		-	-	

664

665 5.4 Summary of proposed tone changes from Proto-Edoid

666 To summarize, we have posited eight tonal correspondence sets for proto-Edoid, divided into three
667 groupings (53). A low grouping consists of **L-L and **L-LH melodies, a high/low grouping
668 consists of **L-H, **L-HL, and **H-L melodies, and a high grouping consists of **H-LH, **H-
669 HL, and **H-H melodies. In the summary table in (53), we notate any changes from Proto-Edoid
670 to the intermediate reconstructions in red, and box any mergers.

671 (53) Proto-Edoid and reflexes in four intermediate branches (changes in red, mergers boxed)

	Proto-Edoid	Proto-DE	Proto-SW	Proto-NC	Proto-NW
a. Low grouping	PE-01 **L-L > *L-L	*L-L	*L-L/*H-L	*L-L	*L-L
	PE-02 **L-LH > *L-L	*L-L	*L-H	*L-H	*L-L
b. High/low grouping	PE-05 **H-L > *L-L	*L-L	*L-H	*H-L	*H-L
	PE-03 **L-H > *L-L	*L-L	*L-LH	*H-L	*H-L
	PE-04 **L-HL > ?		*L-L	*H-L	*H-L
c. High grouping	PE-07 **H-HL > *H-H/*H-L	*H-H/*H-L	*H-L	*H-L	*H-L
	PE-06 **H-LH > *H-H/*H-L	*H-H/*H-L	*H-L	*H-LH	*H-L
	PE-08 **H-H > *H-H/*H-L	*H-H/*H-L	*H-H/*H-L	*H-L/*H-H	*H-L

672

673 Looking across the four branches, we notice largely the same array of diachronic tone changes
674 that we established for the four intermediate branches to modern Edoid languages (§4.5 above).
675 H-Obligatoriness in Proto-Edoid – the addition of a high tone to all-low sequences – is not
676 discernible to the same degree as it is for individual Edoid languages, though we still see its effects
677 in PE **L-L > P-SW *H-L (albeit with an unclear conditioning environment). Another established
678 change, H-Lowering, is more common and is found in all four branches in the high grouping.
679 Further, Rightward H-Shift happened in the development of P-DE and P-SW where the H of **H-
680 L melodies moves into the stem (eventually lost in P-DE), which is blocked from happening if the
681 stem already contains a H tone. This is directly analogous to the conditions of the Great Edo Tone
682 Shift from P-NC *H-L to modern-day Edo L-H (§4.3). Finally, just as we saw rising sequences as
683 marked in SW (called Rising-Markedness), **L-LH too is marked in PE and eliminated in favor
684 of *L-L and *L-H.

685 While these diachronic changes do indeed recur from PE to P-DE/P-SW/P-NC/P-NW and
686 then from these intermediate branches to the modern languages, there are additional diachronic
687 changes which seem to have taken place only in PE. One notable tone change is a “Leftward H-
688 Shift” in P-NC and P-NW which changed PE **L-H and **L-HL into *H-L, and acts as the logical
689 counterpart to the more common Rightward H-Shift. Specifically, Leftward H-Shift only takes

place if the **H is at the stem boundary already – it is not found if it is further away, e.g. **L-LH becomes *L-H/*L-L (not *H-L). Typologically, leftward shift is much rarer than rightward shift, an observation we return to in the discussion section (§6.1).

Another set of new diachronic changes are implied by the development of P-SW, which retains much of the tonal contrast of PE but with key reorganizations. One is Low-Plateauing in which H is lowered between two L's, i.e. PE **L-HL > P-SW *L-L. More complicated is what appears to be the swapping of two values: PE **L-H becomes P-SW *L-LH but PE **L-LH becomes P-SW *L-H. Whether this is the best analysis and how such a swap could come about (likely through a series of changes) remain for future work.

In sum, all branches involve a reduction of the tone classes of PE, especially with respect to multiple tone values on the stem portion of the noun. P-DE collapses the distinction essentially into a *L-L vs. *H-H/*H-L distinction based on the original tone on the prefix (*modulo* the rightward shift of the H in **H-L first into the stem). A similar collapse happened in P-NW, where there is a *L-L vs. *H-L distinction also dependent on the tone on the prefix (again, *modulo* the leftward shift of H in **L-H and **L-HL to the prefix position). In contrast, P-SW and P-NC maintain more original contrasts but by different means. P-SW shifts H rightward in **H-L (but not with other H- initial contexts where the stem H blocks shift), while P-NC shifts the H of **L-LH, **L-H, and **L-HL leftward one position (but not with *H-LH due to the presence of the prefix H). Based on this reconstruction, the most enduring tone preserved from PE is the prefix tone, followed by the stem-initial tone, and finally the stem-final tone.

6 Discussion

Having established our proposal for Proto-Edoid tone, we discuss two issues: interpreting Edoid within the larger typology of tone changes (§6.1) and assessing the regularity of Edoid tone change (§6.2).

6.1 Edoid within the typology of diachronic tone changes

Within each stage in the development Edoid tone, we have posited a specific reconstructed value, e.g. the three P-SW correspondence sets SW-01, SW-02, and SW-03 are reconstructed as *L-L, *L-LH, and *L-H to account for the distribution of L-L and L-H in the modern SW languages (even though no language maintains the *L-LH pattern). This is opposed to an alternative which would simply reconstruct these as *L₁, *L₂, and *L₃ (or *LH₁, *LH₂, and *LH₃, or some combination). By avoided such “diacritics”, for each proto-melody there is a precise diachronic change which can be assessed for typologically plausibility and language-internal coherence. This is important for cross-linguistic work on tone reconstruction, because the typology of tone change sorely lags behind that of segment change, which reduces our collective knowledge on what is common vs. rare vs. unattested (Campbell 2021).

In total, we have posited six types of tone changes when examining all stages in the development of Edoid, summarized in (54).

(54) Diachronic tone changes grouped into six types in all stages of Edoid development

- a. Rightward H-Shift (e.g. H-L > L-H)
- b. Leftward H-Shift (e.g. L-H > H-L)
- c. H-Lowering (e.g. H > M; H-H > H-[↓]H; L-HL > L-L; *etc.*)
- d. L-Raising (e.g. H-L > H-[↑]H or H-M)
- e. H-Obligatoriness (e.g. L-L > H-L or L-H)
- f. Rising-Markedness (e.g. L-LH > L-H or L-L)

The occurrence of these changes is uneven across the various stages of Edoid. This is summarized in (55), which divides PE to the four proto-branches, and then from these intermediate reconstructions to the modern-day languages. If a language demonstrates the tone change, its cell

is marked with an X (which we place in parentheses if the change applies to only a minority subset of the relevant cognates).

(55) Diachronic tone changes by individual language

	PE				DE			SW					NC						NW						
	P-DE	P-SW	P-NC	P-NW	deg	enn	epi	erh	iso	oke	urh	evh	bin	esa	ema	ets	une	aaa	oso	atg	olm	opa	uha	ehu	uku
R'wrd H-Shift (HL > LH)	X	X					X						X												
L'wrd H-Shift (LH > HL)			X	X																					
H-Lowering (H > ⁺ H/M/L)	X	X	X	X	X	X	X	X	X	X	X			X	X	X	X	X	X						X
L-Raising (L > ⁺ H/M)							X				(X)														
H-Obligatory (∅ > H)		(X)			X			X	X				(X)								X				
Rise-Marked (-LH > -L/-H)	X	X	X	X				X	X	X	X	X													

Let us focus first on the top two involving H-shifting. From (55), leftward shift is only hypothesized for P-NC and P-NW to account for correspondences with L-initial cognates in P-DE and P-SW. In contrast, evidence for rightward shift is clearly seen in modern day languages, such as Edo where P-NC *H-L becomes L-H systematically (the Great Edo Tone Shift – §4.3). The prominence of rightward over leftward shift complies with cross-linguistic trends. Perseverative H-shifting or H-spreading rightward is much more common than its counterpart Anticipatory H-shifting/spreading leftward (Hyman & Schuh 1974; Hyman 1978, 2017). This complies with the tendency for tone targets to be realized late, part of the literature on pitch or peak delay (Yip 2002; Kingston 2003; Akinlabi & Liberman 2007; Zhang & Meng 2016:172). Still, there are enough instances of leftward shift cross-linguistically that inferring its appearance in a hypothetical intermediate reconstruction is certainly plausible (see examples catalogued in Hyman 2017 and subsequent discussion). Thus, both rightward and leftward shift took place in the development of

Edoid, even resulting in diachronic oscillation: Proto-Edoid melodies ****L-H** and ***L-HL** become P-NC ***H-L**, which then become L-H in present-day Edo (i.e. ****L-H** > ***H-L** > L-H).⁷

Moreover, operations affecting and involving H tones (such as H-Lowering and H-Obligatoriness) are more common than those with L tones (such as L-Raising). H-Lowering operations are diverse, affecting H as a whole (e.g. ***H** > M in Ghotuo [aaa]), and H after both Hs (e.g. ***H-H** > H-L in Eruwa [erh]) and after Ls (e.g. PE ****L-H** > ***L-L** in P-DE). As are H-Obligatoriness constraints, adding a H to either the left or right edge of the word depending on the language. In contrast, L-Raising essentially took place only twice: ***H-L** > L-HM (after shift) in Epie-Atissa [epi] (assuming an intermediate stage L-HL), and ***H-L** > H-⁺H in Urhobo [urh] (and in this latter case, not without caveats – see §4.2). Still, there are certain changes which involve both L and H at the same time, as in the case of Rising-Markedness which changes ***L-LH** to L-L or L-H in order to avoid this kind of stem-level rise after a prefix L.

6.2 On the regularity of tone change

Our efforts to reconstruct proto-Edoid tone shows both the success and limits of regular sound change as applied to tone. The regularity of sound change principle (or “Neogrammarian Hypothesis” – Crowley & Bowerman 2010:170) states that sound change applies whenever the context is met (e.g. **k** → **c** / __i), which entails that all else being equal a sound change will affect all relevant words in the lexicon (for a recent summary and numerous references, see Labov 2020).

At the same time, there have been several challenges to this principle. Most relevant with respect to this paper, tone in particular shows non-compliance with the regularity principle. In-depth surveys purporting irregularity in tone change go back at least to Cheng & Wang (1977),

⁷ This is in fact directly parallel to the development of Akan [aka] dialects, as detailed in Stewart (1983).

774 who document a lexical split of reflexes of Middle Chinese tone III in the Chinese variety
 775 Chaozhou [nan] without a clear phonetic or phonological condition (see also Labov 1981). More
 776 recently, Dockum (2019:89ff.) summarizes the skepticism towards tone within the historical
 777 linguistics literature. Even in highlighting the success of the “Tonal Comparative Method”,
 778 Dockum ties regular tonal changes directly to segmental origins causing tone splitting, which
 779 implies that without clear tonogenesis events (as in the case of Edoid and most of Africa – §1)
 780 regularity in tone change is obscured. As a whole, because the realization of tone as phonetic pitch
 781 is entirely relative, tone appears ripe for the generation of more surface variants which over time
 782 may result in daughter languages with more irregular tone changes compared to segmental
 783 changes.⁸

784 Within Edoid tone, we have posited subsets for virtually every correspondence set, which
 785 could be interpreted as irregular change. In many cases, only a small number of cognates deviate
 786 from the expected reflex, e.g. in P-DE *L-L remains L-L in Engenni [enn] in 83/86 instances.
 787 Only a small number have an unexpected high tone, reflexes of HAND_ARM ʒbɔ̌, LEAF èbɛ́, and
 788 VEHICLE_BOAT ɔ̌kɔ̌. More serious are cases where the number of exceptional cognates is much
 789 higher. Examples include P-SW *L-L which are evenly split in Eruwa: roughly half remaining L-
 790 L (e.g. WORD_2 èmì and CHARM_DRUG_3 ùmù) and just over half becoming [L-H] (e.g.
 791 WATER àmí and WAR_FIGHT_2 èmú). In parallel, P-NC *L-L regularly remains L-L in Edo
 792 (e.g. TIME èyè) but gains a H tone in approximately 20% of cognates (e.g. CHAIN éyǎ̌).

793 Unexpected outcomes are also recurrent in the proto-melodies involving with H tones. For
 794 example, PE **H-H remains *H-H in most P-DE cognates but becomes *H-L about half as many

⁸ At the same time, there are researchers whose painstaking work demonstrate not only that tone is largely regular, but that it does not change at faster or slower rates compared to segments (Auderset 2024, on the Mixtec family).

795 times (cf. Degema SHEEP_1 $\acute{u}^{\star}s\acute{o}m\acute{a}n < P-DE$ *H-H vs. NOSE $\acute{i}s\acute{u}\beta\grave{e}n < P-DE$ *H-L, both
796 reconstructed as PE **H-H). And in Urhobo, we have said that the reflex of P-SW *H-L has two
797 outcomes with an unclear conditioning environment: it is either retained as H-L ($n=42$) or becomes
798 H-M with H-Raising ($n=27$).

799 Such cases of apparent exceptions to the regularity of sound change (principle) are the noise
800 through which we observe systematic tone change. There are two opposing positions one could
801 take in response to this “noise”. One would be that irregular change is an inevitable conclusion,
802 which would then reverse the burden of explanation to asking why certain changes are so
803 overwhelming regular (e.g. P-DE *L-L becomes surface L-H in virtually all Degema cognates).
804 The second would be to assume regular change in nearly all cases, with apparent irregularities due
805 to hidden phonological environments (whether segmental or tonal), morphological factors (e.g.
806 analogy), or kinds of dialect mixing (in line with the original Neogrammarian Hypothesis).

807 We can zoom in on evaluating these opposing positions for Edoid within the South-West
808 branch where it appears most pressing. The PE classes **L-H (PE-03), **L-HL (PE-04) and **H-
809 L (PE-05) have basically uniform reflexes in DE, NC, and NW branches (in DE, they become *L-
810 L, while in the latter two they become *H-L – refer to the summary table in (53)). It is SW which
811 justifies this three-way distinction, where the reflex is *L-LH, *L-L, and *L-H, respectively. The
812 question is whether this reflects tone class conservation in SW and merger in DE, NC, and NW,
813 or is this class innovation strictly bound to SW, whose answer has serious consequences to the
814 inventory of tone changes in Edoid and its implications for the wider typology of tone change
815 (§6.1).

816 We have assumed SW is conservative here, as there is no apparent phonological or
817 morphological factor which would have conditioned melody splitting, e.g. consonantal features

like fortis/lenis (Maddieson 1974), contrastive prefix tone, or compounding. Perhaps the best candidate is syllable loss and different waves of segmental reduction in the history of Edoid, leading to various kinds of distinct surface tone patterns from the same original tone class.⁹ As it stands, we posit more proto-melodies as a hypothesis to be tested, which could later be collapsed if it were found to be unsupported.

7 Conclusion

The immediate goal of this paper was to reconstruct tone on nouns in the Edoid family of Nigeria, one branch within the Benue-Congo subfamily of the Niger-Congo phylum, the largest family in Africa. African languages are important to studies of tone diachrony as no original tonogenesis event is discernible, and we can thus see how tone patterns develop after tone has fully departed from any segmental origins. To this end, we developed the Proto-Edoid Tone (PEdT) database drawn from 40 sources on 21 modern Edoid languages, totaling over 3000 individual data points, from which 436 cognates can be identified. Based on the PEdT database, we systematically compared tone patterns across the four proposed branches of Edoid which led to a number of tonal correspondences in each branch reconstructed as combinations of tone on a noun class prefix and tone on a nominal stem (i.e. nominal tone melodies). Based on these intermediate reconstructions, we reconstructed tone in Proto-Edoid itself as consisting of the symmetrical set of melodies ****L-L**, ****L-LH**, ****L-H**, ****L-HL**, ****H-L**, ****H-LH**, ****H-HL**, and ****H-H**. From these Proto-Edoid melodies, we tracked several recurring tone changes in Edoid, the most common of which were Rightward H-Shift (e.g. the Great Edo Tone Shift, ***H-L** > **L-H**), H-Obligatoriness (i.e. adding a H to all-L words), and H-Lowering (i.e. ***H** > [↓]**H**, M, or L). These tone changes showed both the

⁹ For similar ideas regarding apparent irregularities in tone correspondence and the role of segmental/syllable loss, see Beam de Azcona (2007:§2) on Southern Zapotec.

hallmarks of regular sound change as well as many unexpected reflexes which cannot be explained at this point.

If these Proto-Edoid reconstructions prove successful, what implications are there for Proto-Benue-Congo and ultimately Proto-Niger-Congo? There exist proposals already reconstructing tone in various Proto-Benue-Congo branches to which Proto-Edoid could be compared, such as Proto-Yoruboid (Akinkugbe 1978), Proto-Igboid (Williamson, Blench, & Ohiri-Aniche 2016), Proto-Lower Cross (Connell 1991) and Proto-Upper Cross (Dimmendaal 1978), and Proto-Grassfields within Bantoid (Elias, Leroy, & Voorhoeve 1984). De Wolf (1971:51) posits *H and *L for Proto-Benue Congo and Hyman (2020) posits the same for Proto-Niger-Congo, though both without explicit correspondence sets. This work on Proto-Edoid is one step closer towards higher-order reconstruction of the largest family in Africa, one of the world's foremost tonal hotspots.

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9 Abbreviations

T = tone; H = high tone; L = low tone; M = mid tone; F = falling; ⁺ = downstep; ATR = Advanced Tongue Root; OCP = Obligatory Contour Principle; U = unspecified; N = noun; DE = Delta Edoid branch; SW = South-West Edoid branch; NC = North-Central Edoid branch; NW = North-West Edoid branch; PE = Proto-Edoid; NC = Niger-Congo; PEdT = Proto-Edoid Tone; P- = proto-

10 References

- 863 Agheyisi, Rebecca N. 1986. *An Edo-English dictionary*. Enugu: Ethiope publishing corporation.
- 864 Agheyisi, Rebecca N. 1990. *A grammar of Edo*. UNESCO.
- 865 Aigbe, E. I. 1985. "Edo-English Dictionary". Unpublished manuscript.
- 866 Akinkugbe, Olufemi Odutayo. 1978. *A comparative phonology of Yoruba dialects, Işẹkiri and*
867 *Igala*. PhD thesis, University of Ibadan.
- 868 Akinlabi, Akinbiyi & Mark Liberman. 2007. "Tonal complexes: The prosodic organization of
869 tones". Rutgers Optimality Archive ROA: 911. <https://roa.rutgers.edu/article/view/941.html>
- 870 Auderset, Sandra. 2024. Rates of change and phylogenetic signal in Mixtec tone. *Language*
871 *Dynamics and Change* 14, 1-41.
- 872 Aziza, Rose O. 1997. *Urhobo tone system*. PhD thesis, University of Ibadan.
- 873 Aziza, Rose O. 2003. Tonal alternations in the Urhobo noun phrase. *Studies in African Linguistics*
874 32(2). 1-24.
- 875 Aziza, Rose O. 2008. Neutralization of contrast in the vowel system of Urhobo. *Studies in African*
876 *Linguistics* 37(1). 1-19.
- 877 Beam de Azcona, Rosemary G. 2013. Problems in Zapotec tone reconstruction. In Antić Zhenya,
878 Charles B. Chang, Claire S. Sandy, & Maziar Toosarvandani (eds.), *Proceedings of the 33rd*
879 *Annual Meeting of the Berkeley Linguistics Society: Special Session on Languages of Mexico*
880 *and Central America*, 3-15. Berkeley: Berkeley Linguistics Society.
- 881 Blanc, Jean-François. 1986. *Le verbe urhobo*. PhD thesis, University of Grenoble.
- 882 Boyeldieu, Pascal. 2000. *Identité tonale et filiation des langues sara-bongo-baguirmiennes*
883 *(Afrique centrale)*. Köln: Rüdiger Köppe Verlag.
- 884 Campbell, Eric. 2021. Why is tone change still poorly understood, and how might documentation
885 of less-studied tone languages help? In Patience Epps, Danny Law, & Na'ama Pat-El (eds.),
886 *Historical linguistics and endangered languages: Exploring diversity in language change*, 15–
887 40. New York: Routledge.
- 888 Cheng, Chin-Chuan, & William S. Wang. 1977. Tone change in Chao-Zhou Chinese: A study in
889 lexical diffusion. In W. S. Wang (ed.), *The lexicon in phonological change*. Berlin: De Gruyter.
890 86-100.
- 891 Connell, Bruce. 1991. *Phonetic aspects of the Lower Cross languages and their implications for*
892 *sound change*. PhD thesis, University of Edinburgh.

- 893 Crowley, Terry & Claire Bowern. 2010. *An Introduction to Historical Linguistics*, 4th Edition.
894 Oxford: OUP.
- 895 de Wolf, Paul P. 1971. *The noun class system of Proto-Benue-Congo*. The Hague: Mouton.
- 896 Dimmendaal, Gerrit J. 1978. *The Consonants of Proto-Upper Cross and their Implications for the*
897 *Classification of the Upper Cross Languages*. PhD thesis, Leiden University.
- 898 Dockum, Rikker. 2019. *The tonal comparative method: Tai tone in historical perspective*. PhD
899 thesis, Yale.
- 900 Donwa-Ifode, Shirley. 1986. Phonetic Variation in Consonants (Isoko). *Anthropological*
901 *Linguistics* 28(2). 149-160.
- 902 Donwa-Ifode, Shirley. 1989. Prefix vowel reduction and loss of noun class distinction: The Edoid
903 case. *Afrika und Übersee* 72. 229-253.
- 904 Egharevba, Jacob. 1968 [1934]. *A short history of Benin*. Ibadan: Ibadan University Press.
- 905 Eigbedion, Genevieve. 1985. *Negation in Esan*. MA Thesis, University of Ibadan.
- 906 Ejele, P.E. 1986. *Transitivity, tense and aspect in Esan*. PhD thesis, University College London.
- 907 Elderkin, E. D. 2008. Proto-Khoe tones in Western Kalahari. In S. Ermisch (ed.), *Khoisan*
908 *Languages and Linguistics: Proceedings of the 2nd international symposium, January 8–12,*
909 *2006, Riezlern/Kleinwalsertal*, 87–136. Köln: Rüdiger Köppe Verlag.
- 910 Elias, Philip, Jacqueline Leroy & Jan Voorhoeve. 1984. Mbam-Nkam or Eastern Grassfields.
911 *Afrika und Übersee* 67. 31-107.
- 912 Elimelech, Baruch. 1976. *A tonal grammar of Etsako*. WPP 35. Los Angeles: Department of
913 Linguistics, UCLA.
- 914 Elugbe, Ben O. 1973. *A comparative Edo phonology*. PhD thesis, University of Ibadan.
- 915 Elugbe, Ben O. 1977. Some implications of low tone raising in southwestern Edoid. *Studies in*
916 *African Linguistics* Supplement 7. 53-62.
- 917 Elugbe, Ben O. 1979. Some tentative historical inferences from comparative Edoid studies.
918 *Kiabàrà* 2(1). 82-101.
- 919 Elugbe, Ben O. 1986. The analysis of falling tones in Ghotuo. In Koen Bogers, Harry van der
920 Hulst, & Marten Mous (eds.), *The phonological representation of suprasegmentals*, 51-62.
921 Dordrecht: Foris.
- 922 Elugbe, Ben O. 1989a. *Comparative Edoid: Phonology and lexicon*. Port Harcourt, Nigeria:
923 University of Port Harcourt Press.

- 924 Elugbe, Ben O. 1989b. The Tone System of Ghotuo. *Cambridge papers in Phonetics and*
 925 *Experimental Linguistics* 4. 1-21.
- 926 Elugbe, Ben O. 1989c. Edoid. In John Bendor-Samuel (ed.), *The Niger-Congo languages: A*
 927 *classification and description of Africa's largest language family*, 291-304. Lanham: University
 928 Press of America.
- 929 Elugbe, Ben O. & Klaus Schubert. 1976. Noun classes and concord in Oloma. *Journal of West*
 930 *African Languages* 1-2. 73-84.
- 931 Emuekpere-Masagbor, G. A, 1997. *Preverbal subject markers in Ivie*. PhD thesis, Sherbrooke.
- 932 Greenberg, Joseph H. 1948. The tonal system of proto-Bantu. *Word* 4. 196-208
- 933 Hammarström, Harald & Forkel, Robert & Haspelmath, Martin & Bank, Sebastian. 2024.
 934 Glottolog 5.0. Leipzig: Max Planck Institute for Evolutionary Anthropology.
 935 <https://doi.org/10.5281/zenodo.10804357> (Available online at <http://glottolog.org>, Accessed on
 936 2024-04-10.)
- 937 Hyman, Larry M. 1974. The great Igbo tone shift. In E. Voeltz (ed.), *Proceedings of the Third*
 938 *Annual Conference on African Linguistics*, 111–126. Bloomington: Indiana University Press.
- 939 Hyman, Larry M. 1978. Historical tonology. In V. Fromkin (Ed.), *Tone: A Linguistic Survey*, 257-
 940 269. New York: Academic Press.
- 941 Hyman, Larry M. 2006. Word-prosodic typology. *Phonology* 23. 225-257.
- 942 Hyman, Larry M. 2017. Synchronic vs. diachronic naturalness: Hyman & Schuh (1974) revisited.
 943 *UC Berkeley PhonLab Annual Report* 13(1).
- 944 Hyman, Larry M. 2020. On Reconstructing Tone in Proto-Niger-Congo. In Valentin Vydrin &
 945 Anastasia Lyakhovich (eds.), *In the hot yellow Africa*, 175-191. St. Petersburg: Nestor-Istoria.
- 946 Hyman, Larry M. & Russell G. Schuh. 1974. Universals of tone rules: Evidence from West Africa.
 947 *Linguistic Inquiry* 5(1). 81-115.
- 948 Hyslop, Gwendolyn. 2022. Toward a typology of tonogenesis: Revising the model. *Australian*
 949 *Journal of Linguistics* 42. 275–299.
- 950 Iweh, Ode. 1983. *La phonologie et le système nominal de l'Urhobo*. PhD thesis, Université
 951 Stendhal (Grenoble 3).
- 952 Kari, Ethelbert E. 2004. *A reference grammar of Degema*. Köln: Rüdiger Köppe Verlag.
- 953 Kari, Ethelbert E. 2008. *Degema-English dictionary with English index*. Tokyo: Research Institute
 954 for Languages and Cultures of Asia and Africa (ILCAA), Tokyo University of Foreign Studies.

- 955 Kingston, John. 2003. Mechanisms of tone reversal. In Shigeki Kaji (ed.), *Cross-linguistic studies*
 956 *of tonal phenomena: Historical development, phonetics of tone, and descriptive studies*, 57-120.
 957 Tokyo: Institute for the Study of Languages and Cultures.
- 958 Klomp, Paul. 1993. *A sketch of the phonology and morphology of Esan*. MA thesis,
 959 Rijksuniversiteit te Leiden.
- 960 Labov, William. 1981. Resolving the Neogrammarian Controversy. *Language* 57(2). 267-308.
- 961 Labov, William. 2020. The regularity of regular sound change. *Language* 96(1). 42-59.
- 962 Ladefoged, Peter. 1968. *A phonetic study of West African languages: An auditory-instrumental*
 963 *survey*. Cambridge: CUP.
- 964 Legbeti, Agnes Temítópé. 2022. *Tone and Aspects of Grammar in Ósòsò, Edo, Nigeria*. PhD
 965 thesis, University of Ibadan.
- 966 Lewis, Ademola Anthony. 2013. *North Edoid relations and roots*. PhD thesis, University of
 967 Ibadan.
- 968 Maddieson, Ian. 1974. A possible new cause of tone-splitting—Evidence from Cama, Yoruba, and
 969 other languages. *Studies in African Linguistics* Supplement 5. 205-221.
- 970 Mafeni, B. O. W. 1969. Isoko. In E. Dunstan (ed.), *Twelve Nigerian languages: A handbook on*
 971 *their sound systems for teachers of English*. London & New York: Longmans; Africana
 972 Publishing.
- 973 Masagbor, Richard. 1989. Noun class and concord in Ivié. *Journal of West African Languages*
 974 19(1). 75-86.
- 975 Moñino, Yves. 1995. *Le Proto-Gbaya: Essai de linguistique comparative historique sur vingt-et-*
 976 *une langues d'Afrique centrale*. Paris: Peeters Publishers.
- 977 Olaoluwa, Ewekeye. 2012. *Aspects of the Phonology of Ososo: An Autosegmental approach*. MA
 978 thesis, University of Ibadan MA thesis.
- 979 Osiruemu, Evarista O. 2005. *Tone and Grammar in Esan*. MA thesis, University of Ibadan.
- 980 Rolle, Nicholas, & Ethelbert E. Kari. 2022. Tone and prosodic recursion in Degema nouns and
 981 noun phrases. *Journal of African Languages and Linguistics* 43(2). 249-283.
- 982 Rolle, Nicholas, Florian Lionnet, & Matt Faytak. 2020. Areal patterns in the vowel systems of the
 983 Macro-Sudan Belt. *Linguistic Typology* 24(1). 113-179.
- 984 Ryder, A. 1969. *Benin and the Europeans 1485-1897*. London: Longman.

- 985 Sæbø, Lilja Maria, Eitan Grossman, & Steven Moran. 2025. Tonogenesis: A diachronic typology.
 986 *Diachronica* 42(3-4). 451-478.
- 987 Schaefer, Ron & Francis Egbokhare. 2007. *Emai dictionary*. Koln: Rudiger Koppe Verlag.
- 988 Schuh, Russell G. 2017. *A Chadic cornucopia*. Oakland, California: California Digital Library.
- 989 Stewart, John. 1983. The Asante Twi tone shift. In I. R. Dihoff (ed.), *Current Approaches to*
 990 *African Linguistics* 1, 235-244. Dordrecht: Foris.
- 991 Thomas, Elaine. 1978. *Grammar Description of the Engenni Language*. Arlington: SIL.
- 992 Thomas, Elaine & Kay Williamson. 1967. "Delta Edoid wordlists: A new version of the
 993 publication". [2009 updated by Roger Blench (ed.); originally published as Occasional
 994 Publication No. 8, Institute of African Studies, University of Ibadan, 1967].
- 995 Ukere, Anthony Obakponovwe. 2005. "Urhobo-English dictionary". Unpublished manuscript.
 996 [Edited by Roger Blench]
- 997 Welmers, William E. 1969. Structural notes on Urhobo. *Journal of West African Languages* 1. 85-
 998 107.
- 999 Williamson, Kay. 1989. Benue-Congo Overview. In John Bendor-Samuel (ed.), *The Niger-Congo*
 1000 *languages: A classification and description of Africa's largest language family*, 247-274.
 1001 Lanham: University Press of America.
- 1002 Williamson, Kay, Roger Blench, & Chinyere Ohiri-Aniche. 2016. "Comparative Igboïd".
 1003 Unpublished Manuscript. Cambridge University.
- 1004 Yip, Moira. 2002. *Tone*. Cambridge: Cambridge University Press.
- 1005 Zhang, Jie, & Yuanliang Meng. 2016. Structure-dependent tone sandhi in real and nonce words in
 1006 Shanghai Wu. *Journal of Phonetics* 54(1). 169-201.

1007 **11 French abstract**

1008 Résumé : L'objectif de cet article est d'appliquer la méthode comparative pour la reconstruction
 1009 du système tonal nominal au sein de la famille édoïde, une branche de la sous-famille Bénoué-
 1010 Congo du phylum Niger-Congo dans laquelle aucun événement de tonogenèse originel n'est
 1011 discernable. À cette fin, nous avons constitué une base de données tonale, nommée *Proto-Edoid*
 1012 *Tone database* (PEdT), compilant des données extraites de 40 sources portant sur 21 langues
 1013 édoïdes modernes et totalisant plus de 3000 items individuels, parmi lesquels 436 cognats ont pu

1014 être identifiés. Sur cette base, nous comparons systématiquement les schèmes tonals attestés dans
1015 les quatre branches de l'édoïde, et identifions un certain nombre de correspondances tonales que
1016 nous reconstruisons sous la forme de combinaisons d'un ton porté par un préfixe de classe nominale
1017 et d'un ton porté par un radical nominal, appelées « mélodies tonales nominales ». À partir de ces
1018 reconstructions intermédiaires, nous reconstruisons le système tonal nominal du proto-édoïde
1019 comme un ensemble symétrique de mélodies : **B-B, **B-BH, **B-H, **B-HB, **H-B, **H-
1020 BH, **H-HB et **H-H. À partir de ces mélodies proto-édoïdes, nous identifions plusieurs
1021 changements tonals récurrents, dont les plus fréquents sont le décalage du ton Haut vers la droite
1022 (par ex. le « Grand Décalage Tonal de l'Edo », *H-B > B-H), l'insertion obligatoire du ton Haut
1023 (c'est-à-dire l'ajout d'un ton Haut aux mots ne comportant que des tons Bas) et l'abaissement du
1024 ton Haut (c'est-à-dire *H > [↓]H, M ou B). Ces changements tonals présentent les caractéristiques
1025 typiques des changements phonétiques réguliers, malgré la présence de nombreux réflexes qui ne
1026 peuvent être expliqués à ce stade, ouvrant ainsi la voie à de futurs travaux sur la tonologie édoïde
1027 et ses liens avec le proto-Bénoué-Congo, et, *in fine*, avec le proto-Niger-Congo.

1028 **12 German abstract**

1029 Zusammenfassung: Das unmittelbare Ziel dieser Arbeit besteht darin, unter Anwendung der
1030 vergleichenden Methode die Nominaltöne innerhalb der Edoid-Sprachfamilie zu rekonstruieren –
1031 einem Zweig der Benue-Kongo-Unterfamilie des Niger-Kongo-Phylums, einer Sprachgruppe, für
1032 die bisher kein ursprünglicher Prozess der Tonogenese bekannt ist. Zu diesem Zweck haben wir
1033 die Datenbank *Proto-Edoid Tone* (PEdT) erstellt, die auf 40 Quellen zu 21 modernen Edoid-
1034 Sprachen basiert und insgesamt über 3000 einzelne Datenpunkte umfasst, aus denen sich 436
1035 Kognaten identifizieren lassen. Auf der Grundlage der PEdT-Datenbank vergleichen wir
1036 systematisch die Tonmuster der vier postulierten Zweige des Edoid; dies führt zur Identifizierung
1037 einer Reihe von Tonentsprechungen, die wir als Kombinationen aus dem Ton eines

1038 Nominalklassenpräfixes und dem Ton eines Nominalstammes rekonstruieren. Wir bezeichnen
1039 diese als *nominale Tonmelodien*. Basierend auf diesen Zwischenrekonstruktionen
1040 rekonstruieren wir das Tonsystem des Proto-Edoid als ein symmetrisches Set von Melodien: **T-
1041 T, **T-TH, **T-H, **T-HT, **H-T, **H-TH, **H-HT und **H-H. Ausgehend von diesen Proto-
1042 Edoid-Melodien zeichnen wir mehrere wiederkehrende Tonveränderungen nach; zu den
1043 häufigsten zählen die *Rechtsverschiebung von H* (z. B. die „Große Edo-Tonverschiebung“: *H-
1044 T > T-H), die *H-Pflicht* (d. h. das Hinzufügen eines H-Tons zu Wörtern, die ursprünglich
1045 ausschließlich aus T-Tönen bestanden) sowie die *H-Absenkung* (d. h. *H > ⁺H, M oder T).
1046 Diese Tonveränderungen weisen sowohl die typischen Merkmale regelmäßigen Lautwandels auf
1047 als auch zahlreiche unerwartete Reflexe, die sich zum gegenwärtigen Zeitpunkt noch nicht erklären
1048 lassen; damit wird der Grundstein für künftige Forschungsarbeiten zum Edoid-Tonsystem gelegt
1049 – insbesondere im Hinblick auf dessen Verhältnis zum Proto-Benue-Kongo und letztlich zum
1050 Proto-Niger-Kongo.

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